CIS-Δ: A single-cell benchmark for predicting the expression response to cis-regulatory perturbation

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Running title: Benchmarking sequence oracles on endogenous single-cell cis-Δ

Abstract

Sequence-to-expression models ("oracles") such as Enformer and Borzoi, and the growing family of generative DNA designers built on them, aim to predict or engineer how a stretch of cis-regulatory DNA sets a gene's expression level. Yet these methods are almost never evaluated on the quantity most relevant to regulatory therapeutics: the **change in expression (Δ)** that a real cis-regulatory perturbation produces in its native genomic context, measured at single-cell resolution. Generative designers are typically scored by the oracle that guided their design (circular), or on episomal reporters stripped of native chromatin. We introduce **CIS-Δ**, a benchmark *resource* that scores a method's predicted Δ for a (cell context, cis-sequence intervention) pair against endogenous single-cell ground truth. Scoring uses a five-axis metric package (direction, magnitude calibration, single-cell distribution match, target specificity, and distal-vs-proximal breakdown) over a leakage-controlled split that holds out both enhancers and genes. CIS-Δ is organized along the **perturbation-response** dimension, the signed, quantitative Δ produced by a defined intervention, and its breadth comes from axis diversity (CRISPRi repression, CRISPRa activation, single-base reporter variant effects (MPRA), natural variation, and cross-screen concordance under one Δ metric) rather than dataset count. We instantiate v1 on the Gasperini et al. (2019) K562 CRISPRi enhancer screen (GSE120861; 90,955 enhancer–gene pairs with measured effect sizes). We extend it to five independent ground-truth modalities: a second CRISPRi screen across four non-K562 cell types, a single-cell CRISPRa activation screen, and two natural-variation eQTL cohorts (GEUVADIS LCL and GTEx whole blood, both SuSiE fine-mapped). Against these we benchmark a distance-to-TSS baseline, the two field-standard enhancer–gene link models (ABC and ENCODE-rE2G), and a panel of sequence oracles spanning the field's architectural diversity: Borzoi and Enformer (convolutional coverage models), AlphaGenome (the 2026 Nature state of the art), Evo2 (an autoregressive genomic language model), and scooby (the first single-cell-decoder oracle). Each oracle is run as an element-ablation or variant-effect predictor. Discrimination estimates carry bootstrap confidence intervals, and the sequence-oracle-versus-baseline comparisons are tested with DeLong's test corrected for multiplicity across the comparison family. The central finding is a coverage-and-ranking gap that recurs across every model architecture we tested (enumerated below): **on the distal enhancers a model can see, it recovers the direction of the perturbation, but (i) most real regulatory links lie beyond the model's receptive field (71% for Borzoi, 88% for Enformer), (ii) aggregate performance collapses to chance once**

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promoter-proximal controls are included, and (iii) a naive genomic-distance baseline outperforms both oracles overall. We show this gap **replicates on an independent second dataset in four other cell types** (the EngreitzLab non-K562 CRISPRi benchmark, GRCh38): receptive-field blindness and distance-beats-oracle hold in every line. This indicates the gap is a property of the sequence-oracle approach, not of one screen. Running the oracle with **cell-type-matched readout tracks** recovers the perturbation-direction signal (sign agreement 0.58 $\rightarrow$ 0.79) but leaves both headline limitations untouched: most links stay out of the receptive field, and a distance prior still out-ranks the matched oracle. This isolates what is fixable (readout) from what is architectural (context length and calibration). A third dataset of the opposite sign, a single-cell CRISPRa screen, extends the benchmark to **up-regulation** (61 activation hits, all positive Δ) and closes the repression-only gap that matters most for regulatory therapeutics. On the promoter-activation subset, which is in-window by construction and so isolates calibration from receptive field, the first up-regulation leaderboard finds the oracle recovers activation direction in neurons (0.80, matching the 0.79 CRISPRi in-window figure) but not in K562 (0.00): in-window visibility is necessary but not sufficient for calibration. A fourth modality, **natural genetic variation** (1,826 fine-mapped GEUVADIS LCL cis-eQTL lead SNVs, eQTL Catalogue, hg38) scored in ref-vs-alt variant-effect mode against GM12878 tracks, shows the finding is not an artifact of engineered perturbations. The oracle recovers eQTL direction only near the TSS (0–5 kb sign agreement 0.65, ρ 0.39, decaying to chance beyond 5 kb). It edges a distance baseline (AUROC 0.595 vs 0.566) but is out-ranked by the fine-mapping posterior itself (0.69), and, unlike distal CRISPRi enhancers, 96% of fine-mapped leads are in-window, so the limit is the oracle's short *effective* range, not coverage. Separately, using the full 207,324-cell matrix, we show the single-cell distribution axis is real and measurable: for the 20 strongest enhancer hits, the perturbed-vs-control per-cell energy distance exceeds a sampling null in 20/20 cases (median 3.2$\times$). Beyond the forward (prediction) task, we build and release the **Track B** generative-design scoring harness: a designer emits a sequence to hit a target Δ and is scored by an oracle walled off from design time. The v1 baselines (random/shuffle/reference) establish a null floor and a reference ceiling, and show the walled-off oracle cannot yet certify a designed sequence at the element level (discrimination AUROC 0.53), the same calibration limit seen from the design side. Finally, a lightweight recalibration head fusing oracle-Δ with a distance prior on held-out enhancers adds a small but real increment over distance alone (AUROC 0.868 $\rightarrow$ 0.876), and quantifies how much of the gap is closable. The finding recurs across the architectures we tested, and the direction of the gap depends on the assay. On K562 CRISPRi the field-standard ABC/rE2G models and a plain distance prior out-rank the sequence oracles; on natural-variation eQTL the picture flips for the 2026 state of the art. Across the full BH-corrected family of 37 oracle-vs-baseline comparisons, 8 are significant: 4 favor a baseline (Borzoi and Enformer below distance/ABC/rE2G on CRISPRi), 2 favor **AlphaGenome over the distance prior** on eQTL (GTEx q<0.001, GEUVADIS q=0.03), and 2 favor the outcome-informed fine-mapping PIP reference over the oracle. So the 2026 state of the art does clear the sequence-agnostic baseline on eQTL magnitude, the first oracle to do so, with the strongest direction and rank recovery of any sequence oracle, yet statistical fine-mapping still out-ranks it, and a genomic language model (Evo2) sits at chance. The one suggestive exception points to the frontier: scooby, built for single-cell output, shows a not-yet-significant trend (Spearman 0.47, p=0.088) on the single-cell distribution axis that coverage oracles cannot address at all. A fifth axis fixes the intervention and varies the *cell context*, the term the virtual-cell framing is named for, using a four-tissue GTEx eQTL panel read under each tissue's matched RNA tracks. On this panel the matched-track advantage is small and uncertain (AUROC 0.595 vs mismatched 0.592; tissue top-1 0.48 vs chance 0.44), but the ground truth is near-degenerate (cross-tissue β correlated 0.93–0.96), so this reads as a benchmark-axis identifiability limit rather than a demonstrated model failure. We reproduce the AlphaGenome result against DeepMind's **official** served model (r=0.990 vs the PyTorch port; ranking unchanged), so the ranking reflects the true model. We ground the benchmark in a therapeutic worked case, the BCL11A fetal-hemoglobin locus targeted by the approved Casgevy therapy. CIS-Δ is provided as working **living-benchmark infrastructure**: a documented submission format spanning **seven scoring axes** (the five-axis metric package plus two perturbation-response tracks, bidirectional single-base reporter MPRA and GWAS-variant single-cell CRISPRi, both provisional/ exploratory), a scoring harness whose `--selftest` reproduces the committed per-axis benchmark scores from the committed data to <1e-6 on a clean checkout across all seven axes, a frozen ground-truth bundle, and an auto-generated leaderboard, so any new model can be scored against the same baselines and the calibration gap tracked as the field advances. We are explicit that this is a **provisional v1 evaluation, not an open benchmark release**: the reference datasets are still under formal admission review, and public release of the leaderboard as a scientific resource is gated on an independent go/no-go decision (Gate A/Gate C) that no analysis in this paper pre-empts.

1. Introduction

A gene's coding sequence specifies *what* protein is made; its cis-regulatory elements (promoters and enhancers) specify *how much*, *where*, and *when*. Most disease-associated common variants identified by genome-wide association studies fall outside coding sequence, within these regulatory elements. The decisive quantity, both therapeutically and biologically, is therefore not whether a gene is on or off but the **fold-change**: by how much a given sequence turns a gene's expression up or down, and in which cell type.

This quantity is already a drug. Casgevy (exagamglogene autotemcel), the first FDA-approved CRISPR therapy (December 2023) for sickle-cell disease and β-thalassemia, works by editing the **erythroid-specific enhancer of BCL11A**. BCL11A represses fetal hemoglobin; disabling its enhancer *only in the red-cell lineage* reactivates fetal hemoglobin while leaving BCL11A intact in every other tissue. The therapeutic logic is to "produce a specific fold-change in a specific gene, in a specific cell type, by changing a cis-regulatory sequence", which is the (context, cis-intervention) $\rightarrow \Delta$ prediction problem studied here.

Two research communities have formed around parts of this problem, but they have never shared a benchmark:

- **Single-cell trans-perturbation prediction** (e.g. the Arc Virtual Cell Challenge) has a mature benchmark culture, scoring how a *gene knockdown* reshapes the transcriptome. Its intervention, however, is a trans genetic perturbation, not a change to regulatory DNA.
- **Generative cis-regulatory design** (DNA-Diffusion, discrete-diffusion designers, expression-optimizing GANs and RL methods, cell-type-specific enhancer/AAV-promoter designers) generates DNA meant to drive a target expression level. But each paper evaluates differently, almost always via **oracle circularity** (scoring the design with the same model that guided it) or **episomal reporters** (bulk MPRA/STARR-seq on a plasmid, without native chromatin or single-cell heterogeneity).

The gap. No shared benchmark scores the single-cell expression *response change (Δ)* caused by a cis-regulatory sequence, real or generated, against endogenous, single-cell ground truth. This is the regime in which sequence oracles are documented to be weakest: they predict expression well *across genomic loci* but poorly *across small sequence edits in a fixed locus*, which is the regime a cis-perturbation benchmark stresses.

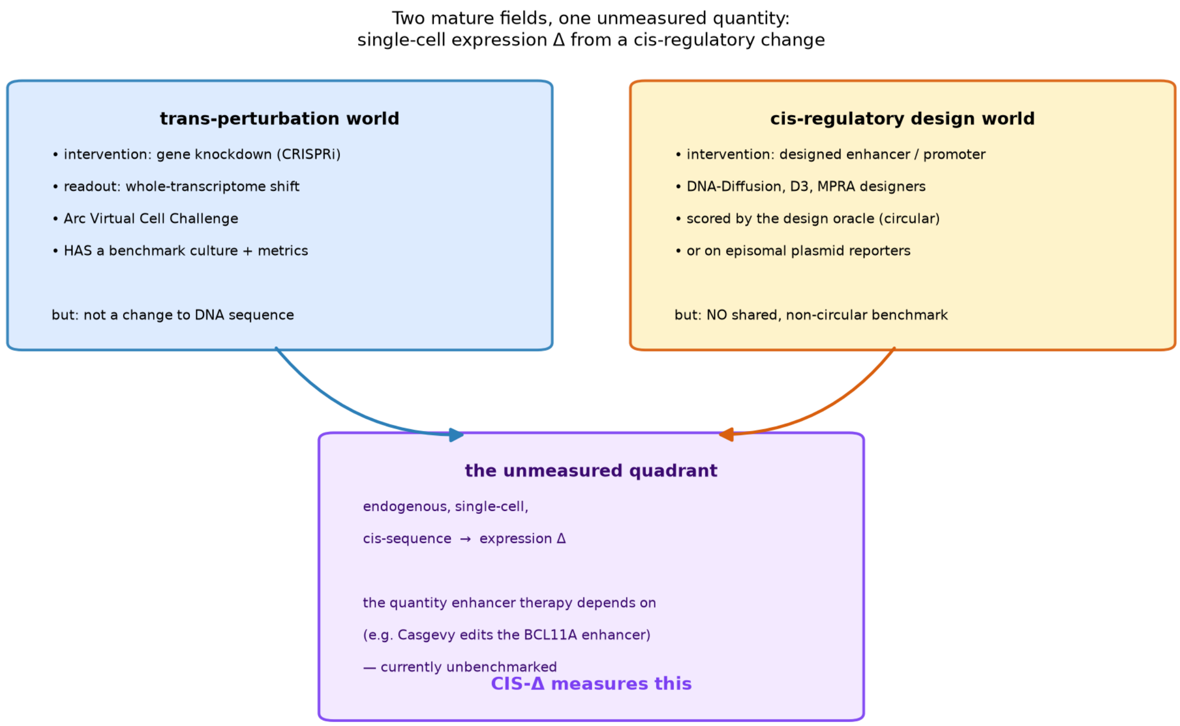


Figure 1 — The gap. Two mature fields (single-cell trans-perturbation prediction; generative cis-regulatory design) leave one quantity unmeasured: endogenous single-cell expression Δ from a cis-regulatory change.

Contributions. We are explicit up front about what is and is not new here. Scoring sequence oracles for target-gene Δ on the Gasperini/Fulco CRISPRi screens overlaps concurrent 2023–2026 work (§2, and the Limitations); the base observation that oracles under-rank a distance prior is not, on its own, our claim to novelty. Our contribution is the packaging that turns that observation into a reusable, falsifiable resource:

1. **CIS-Δ**, a benchmark that scores cis-sequence $\to$ single-cell Δ against endogenous ground truth under a single unified framing, with a two-track design (forward: predict the Δ of a real perturbation; inverse: design a sequence to hit a target Δ, scored by an oracle walled off from the design oracle) and axes spanning repression, activation, and natural variation.
2. A **five-axis metric package** (`cisdelta.score`) that prevents a model from winning by predicting only the mean response.
3. A **leakage-controlled evaluation** holding out both enhancers and genes, with a frozen split for cross-submission comparability.
4. A **reference evaluation** of a distance baseline and two published oracles (Borzoi, Enformer) that surfaces a concrete, reproducible calibration gap, plus a **recalibration head** that quantifies how much of the gap is closable by fusing oracle signal with a distance prior on held-out endogenous Δ.

2. Related work

Sequence-to-expression oracles. Enformer and Borzoi predict thousands of functional genomic tracks (CAGE, RNA-seq, DNase, ChIP) from DNA sequence over long context windows (~196 kb and ~524 kb respectively). They are widely used off-label as variant- and element-effect predictors by comparing predictions for reference vs. altered sequence. scooby extends this to single-cell-resolution profiles; AlphaGenome improves variant-effect prediction but retains the across-individual limitation. A distinct paradigm is the autoregressive genomic language model, such as Evo2 (StripedHyena2), trained by next-nucleotide prediction with no functional-genomics supervision. Such models score variants zero-shot by change in sequence likelihood; we include one to test whether the calibration gap is specific to the coverage-prediction inductive bias.

Enhancer–gene link prediction. The Activity-By-Contact (ABC) model and supervised successors (ENCODE-rE2G) predict which enhancers regulate which genes from accessibility and 3D contact, and provide strong non-sequence baselines.

Perturbation benchmarks. The Arc Virtual Cell Challenge showed that even well-resourced models struggle to beat naive baselines at trans-perturbation prediction; we carry that lesson over by including a distance baseline as a mandatory floor.

Positioning. CIS-Δ scores, at single-cell resolution, the **signed, quantitative expression response (Δ) to a defined cis-regulatory *perturbation***: CRISPRi repression, CRISPRa activation, single-base reporter variant effects (MPRA), and natural variants. Its contribution is breadth *along the perturbation-response axis*: many perturbation modalities scored under one Δ metric, rather than a larger count of datasets.

Generative regulatory design. DNA-Diffusion, discrete-diffusion designers, and lentiMPRA-validated synthetic-enhancer libraries generate cis sequences with target activity, but lack a shared, non-circular evaluation, the gap that CIS-Δ's Track B is built to fill.

3. The CIS-Δ benchmark

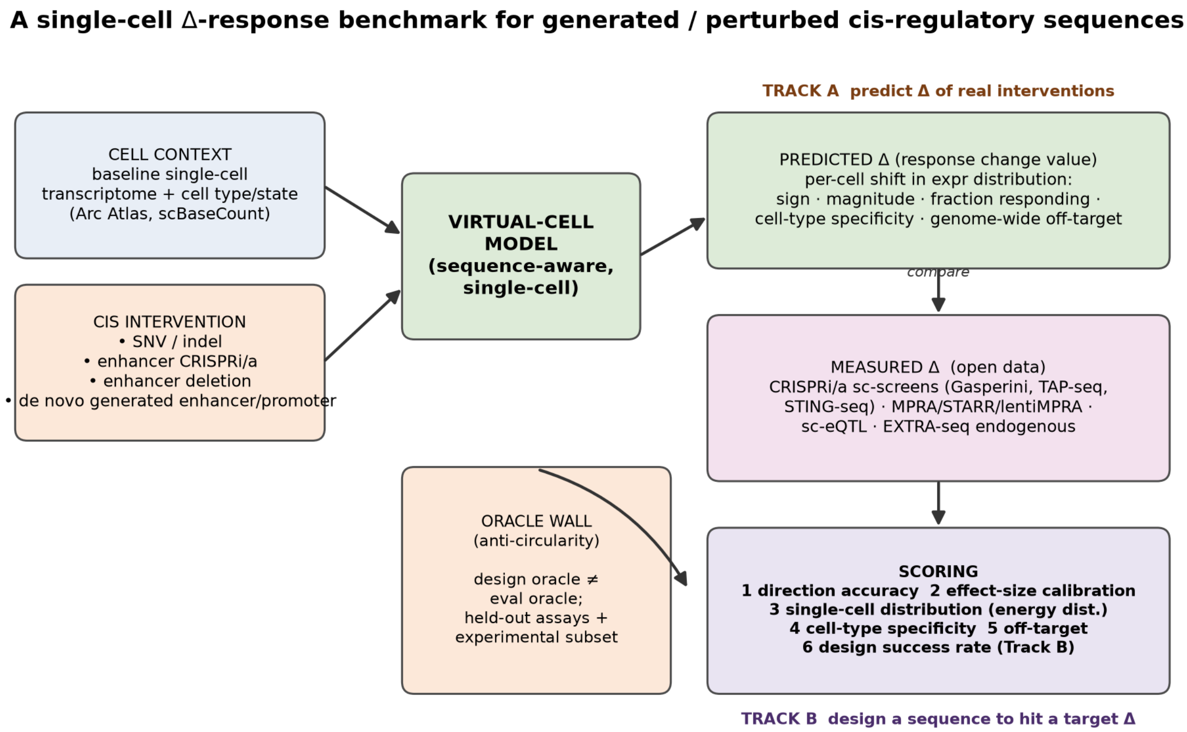


Figure 2 — The benchmark. The CIS-Δ concept: a (cell context, cis-intervention) → Δ unit, two tracks (forward prediction /inverse design), a five-axis metric package (plus two perturbation-response tracks, §5.21–5.22), and the anti-circularity oracle wall.

3.1 Unit of evaluation

The evaluation unit is a **(cell context, cis-sequence intervention)** → Δ triple. A method receives a baseline single-cell context and a defined cis change (SNV, enhancer CRISPRi/CRISPRa, or enhancer deletion) and must predict the shift in target-gene expression. Δ, not the raw expression level, is the scored quantity. Δ is what matters for both therapeutics and mechanism, and the same edit produces a *distribution* of responses across cells that a mean-only prediction cannot capture.

3.2 Two tracks

- **Track A (forward /discriminative).** Predict the Δ of a real intervention. Ground truth: single-cell CRISPRi/CRISPRa screens, sc-eQTL, MPRA.
- **Track B (inverse /generative).** Given a target cell type and desired Δ, emit a DNA sequence, scored by an **independent oracle explicitly walled off from the design oracle**, plus an experimental readout on the top-k designs. Track B's scoring harness and baselines are now implemented (§5.10); a real generative model is the remaining plug-in.

3.3 The scoring axes

CIS-Δ is organized as a set of **evaluation axes**, each a distinct perturbation-response modality scored under one common Δ metric. Five axes form the metric package applied to every dataset (below). Two further **perturbation-response tracks** (§5.21–5.22) extend the resource to modalities the anchor could not test: bidirectional single-base editing and GWAS-variant single-cell CRISPRi. This axis-diversity, not dataset count, is the resource's defining property. For each (element, gene) pair a method emits a predicted Δ^; ground truth is the measured effect size β (and, where available, a per-cell Δ distribution):

1. **Direction /AUROC** — sign(Δ^) vs sign(β) on significant pairs; AUROC of |Δ^| ranking hits vs. matched non-hits.
2. **Magnitude calibration** — regression of β on Δ^: slope (1.0 = calibrated), intercept, dispersion, and Spearman rank; a calibration curve. (Deliberately not reduced to a single Pearson r.)
3. **Single-cell distribution match** — energy distance /MMD between predicted and observed per-cell Δ distributions, calibrated against an NTC–NTC null. *This is the axis no existing benchmark provides.*
4. **Target specificity** — each enhancer is tested against its true target and several nearby non-targets; does the method concentrate Δ^ on the true target? Reported as specificity AUPRC.
5. **Distal vs. proximal** — scored separately on distal enhancer links vs. TSS-proximal controls; the hypothesis is that oracles collapse on distal.

The package is released as `cisdelta.score(pred_df, truth_df)` $\to$ `dict` against a frozen, leakage-controlled split.

3.4 The seven scoring axes at a glance

CIS-Δ's breadth is the diversity of *perturbation modalities* it scores under one Δ metric, not the number of datasets. Table 1 lists the seven scoring axes the harness currently supports (`code/score_submission.py --axis`), spanning CRISPRi repression, CRISPRa activation, single-base reporter MPRA, and natural variation, the perturbation-response dimension. Every axis is a **provisional v1 fixture**: none is V4-admitted, and model scoring on any axis unlocks only after the scoring SAP is signed (§3.2, Methods).

Table 1 — The seven CIS-Δ scoring axes. Independent unit N is the generalization unit (loci, fine-mapped variants, elements, or genes), not the raw row count. The effect-sign column marks which axes can test direction: the CRISPRi axes are repressive-only (direction-degenerate on the distal-DHS stratum), CRISPRa is activating-only, `satmpra` is the one **bidirectional** reporter-MPRA axis, and the eQTL axes carry both signs by construction.

Axis (`--axis`)	Dataset (assay)	Perturbation modality	Effect sign	Independent unit (N)	Primary metric
`gasperini_dhs`	Gasperini 2019 K562 CRISPRi enhancer ablation (hg19)	CRISPRi repression	repressive	~30 distal-DHS hit loci (3,789-pair test split; 3,513 distal-DHS)	`direction_auroc_dhs`

Axis (`--axis`)	Dataset (assay)	Perturbation modality	Effect sign	Independent unit (N)	Primary metric
`eqtl_geuvadis`	GEUVADIS LCL eQTL (QTD000110, SuSiE, hg38)	natural variation	bidirectional	1,826 fine-mapped lead variants	`auroc_magnitude`
`eqtl_gtex`	GTEx v10 whole-blood eQTL (QTD000356, SuSiE, hg38)	natural variation	bidirectional	1,800 fine-mapped lead variants	`auroc_magnitude`
`multitissue`	GTEx v10 4-tissue eQTL (blood/brain/muscle/skin, hg38)	natural variation (cell-type)	bidirectional	5,619 pairs fine-mapped in $\geq$1 tissue	`matched_minus_mismatched`
`promoter_crispra`	Chardon 2024 CRISPRa promoter activation (K562+neuron, hg38)	CRISPRa activation	activating	18 loci (10 responsive)	`locus_dir_agreement_responsive`
`satmpra` *(provisional)*	Kircher/Ahituv saturation-mutagenesis MPRA (GSE126550)	reporter variant effect (episomal MPRA, single-base)	**bidirectional**	21 elements (7,324 sig. edits)	`sign_agreement_significant`
`sting_seq` *(provisional)*	STING-seq GWAS single-cell CRISPRi (GSE171452, K562)	CRISPRi repression (GWAS cCRE)	repressive	124 genes /30 concordant w/ anchor	`sign_agreement`

(”Provisional” and ”exploratory tier” are used synonymously across the project’s documents: both mean a fixture that failed a V4 admission gate (satMPRA on power, STING-seq on independent-validation overlap) and is retained for completeness, not as an admitted benchmark axis.)

All seven round-trip in the scoring harness’s `--selftest` to <1e-6 from committed data; the two perturbation-response tracks (`satmpra`, `sting_seq`) are detailed in §5.21–5.22.

4. Reference dataset and construction (v1)

4.1 Anchor dataset

We anchor v1 on **Gasperini et al. (2019), GSE120861**, a genome-scale K562 CRISPRi enhancer screen ("A genome-wide framework for mapping gene regulation via cellular genetic screens," *Cell* 2019; verified live against NCBI, *Homo sapiens*, 53 samples, genome build hg19/GRCh37). Its precomputed differential-expression table provides, per enhancer–gene pair, the effect size β, adjusted empirical p-value, enhancer coordinates (`target_site.*`), target-gene TSS (`target_gene.*`), and a `site_type` label (DHS candidate enhancer /TSS /selfTSS /NTC control /positive control). Each significant link is thus a ready-made **(DNA element with coordinates, target gene, measured Δ)** triple.

For cross-cell-type replication (§5.6) we add a **second dataset**: the EngreitzLab curated CRISPRi enhancer–gene benchmark (GRCh38), restricted to its non-K562 lines (WTC11, HCT116, Jurkat, GM12878). This second dataset contributes 2,460 tested pairs and 103 significant distal links in the same (element, gene, effect-size, significance) schema. It is coerced into the identical benchmark table format by `09_build_engreitz_table.py`, so it feeds the same oracle and scoring code unchanged.

4.2 Benchmark table (verified from the file)

Quantity	**Value**
Total tested pairs (full DEG file)	629,515
NTC control pairs (dropped)	538,560
Benchmark pairs (DHS + TSS + selfTSS + positive_ctrl)	**90,955**
DHS candidate-enhancer pairs	84,595
Unique candidate enhancers (DHS)	6,100 (5,723 distinct loci)
Enhancer element width	median 417 bp per unique locus (448 bp over all rows)
Unique target genes tested	10,560
Significant links (padj < 0.1)	**1,306** (all repressive)
Significant distal DHS enhancer$\rightarrow$gene links	808 (719 enhancers, 513 genes)
Median genes tested per enhancer	10 (max 72)
Effect size β (significant)	median ≈ -0.35, range to -3.69

All significant effects are repressive ($\beta < 0$), consistent with dCas9-KRAB CRISPRi. We pin the sign convention directly from the data: across 369 selfTSS positive-control hits, 100% have $\beta < 0$ (median -0.77), fixing "negative β = repression."

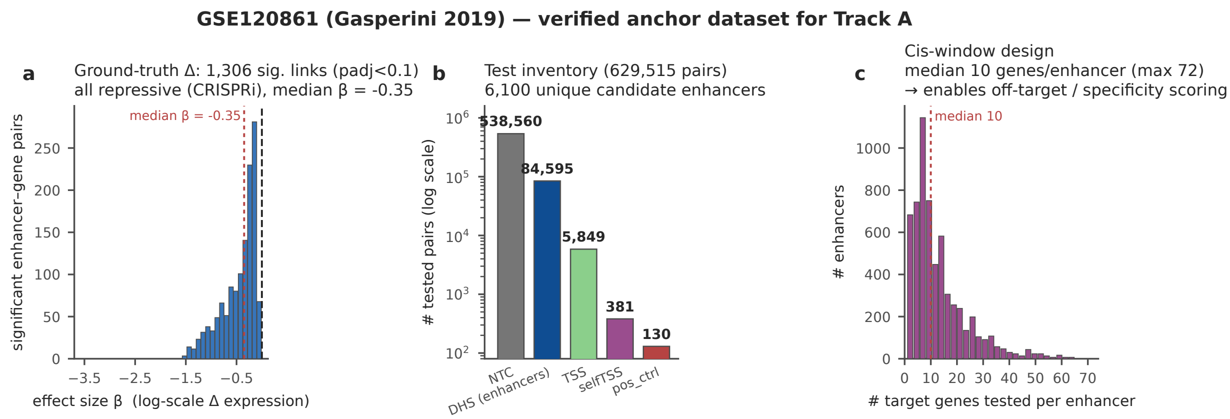


Figure 3 — The anchor dataset is real and rich. Verified structure of Gasperini GSE120861: effect-size (β) distribution of significant links, test inventory, and genes-per-enhancer (the built-in specificity test).

4.3 Sequence extraction, splits, baselines

Enhancer DNA is pulled directly from an **hg19 FASTA** at the element coordinates. No liftOver is required, because a sequence oracle consumes DNA, and the DNA is identical regardless of coordinate label. liftOver is needed only to join elements to hg38-coordinate epigenomic tracks. We extract 6,491 element windows (flank 250 bp); only one window contained an ambiguous base.

The **frozen split** holds out both enhancers and genes: the test set (3,789 pairs) shares zero enhancers and zero genes with the training set, so a method cannot succeed by memorizing either. Baselines: **distance-to-TSS** (a naive floor, per the Arc VCC philosophy) and a **mean-Δ constant** predictor.

4.4 Storage and compute

The ground-truth-only benchmark is ~83 MB gzipped (≈500 MB unzipped), which is laptop scale. Adding the per-cell distribution axis requires the 8.9 GB single-cell matrix (~25–30 GB unzipped). Oracle inference over the test split runs in minutes-to-an-hour on a single A100. The complete v1 working set is ~15 GB.

5. Reference oracle evaluation

5.1 Oracles as element-ablation predictors

We score each published oracle by **element ablation**. We center the model's input window on the target-gene TSS (hg19) and predict the K562 expression signal at the TSS output bin, summed over a ±5-bin window and over all matched K562 tracks. We then replace the enhancer element in place with a composition-preserving shuffle, which destroys motifs while preserving GC content, and re-predict. The predicted Δ is the log2 change in TSS signal. A real activating enhancer, when ablated, lowers predicted expression (Δ^ < 0), matching the repressive CRISPRi convention. Reference passes are cached per unique TSS. Elements beyond the model's receptive field are recorded as Δ^ = NaN with `window_ok = False`, and reported rather than silently dropped.

- **Borzoi** (`johahi/borzoi-replicate-0`): 524,288 bp input, 6,144 output bins (32 bp), 94 matched K562 RNA tracks; half-window ≈ 262 kb.

- **Enformer** (`EleutherAI/enformer-official-rough`): 196,608 bp input, 896 output bins (128 bp), K562 CAGE tracks; half-window $\approx$ 98 kb.

5.2 Leaderboard

The whole-test-set leaderboard scores every method on the full 3,789-pair test split (out-of-window oracle predictions treated as no-Δ, as they would be for any real submission that must return a value for every pair):

Method	Direction AUROC (all test)	Calibration slope	Spearman	Distal AUROC	Proximal AUROC
distance (naive floor)	**0.868**	0.394	0.441	0.804	0.950
mean-Δ constant	0.500	—	—	0.500	0.500
Borzoi (ablation)	0.493	0.097	−0.223	0.873	0.500
Enformer (ablation)	0.368	0.11	−0.18	0.724	0.500

Restricting each oracle to the pairs it can actually see (in-window) tells the more diagnostic story:

Oracle	Half-window	In-window frac	AUROC (all in-window)	AUROC (DHS in-window)	DHS sign-agree	DHS Spearman(β)
Borzoi	262 kb	29.3% (1,109)	0.493	**0.873**	1.00	+0.27
Enformer	98 kb	12.0% (453)	0.368	**0.724**	0.94	+0.20

The naive distance baseline (AUROC 0.868) beats both oracles on the full test set.

5.3 The finding

Three results define the calibration gap, and, critically, **all three hold for both oracles**, so the gap is a property of the sequence-oracle approach on this task, not of one model:

1. **Receptive-field blindness.** Only 29% of test pairs (1,109/3,789) fall within Borzoi's 262 kb half-window, and only 12% (453/3,789) within Enformer's 98 kb half-window. Thus **71% and 88% of tested regulatory links, respectively, lie beyond the model's context entirely**. Across the 808 significant distal DHS links, enhancers act at a median 43 kb from their target but out to ~995 kb (the significant hits in the held-out test split sit further still, median ~85 kb). A model structurally cannot score a link it cannot see, and this is the single largest source of the gap.

2. **Direction is recovered on visible distal enhancers.** On in-window DHS pairs, Borzoi ranks CRISPRi hits from non-hits at **|pred|-AUROC 0.873** (sign agreement 1.00, Spearman β +0.27) and Enformer at **0.724** (sign agreement 0.94, Spearman +0.20). The biology is partially captured by both: ablating a true enhancer lowers predicted expression.

3. **Promoter controls collapse the aggregate, and the naive baseline wins.** Mixing promoter-proximal TSS/selfTSS controls back in drops the aggregate in-window AUROC to **0.493 (Borzoi)** and **0.368 (Enformer)**, at or below chance, because proximal ablation behaves erratically. On the full test set the **distance baseline (AUROC 0.868) beats both oracles**, echoing the Arc VCC lesson that naive methods are hard to beat. Enformer, with the shorter window and only FANTOM CAGE (not RNA-seq) K562 tracks available, is uniformly weaker than Borzoi. The gap does not close with a second model; it widens.

Both oracles show the same signature: mostly-zero predictions for promoter controls (whose 1-bp point "elements" shuffle to a no-op), a near-flat calibration slope (0.10 Borzoi, 0.11 Enformer, vs. 1.0 for a calibrated model), and directional signal that survives only on the distal enhancers within range.

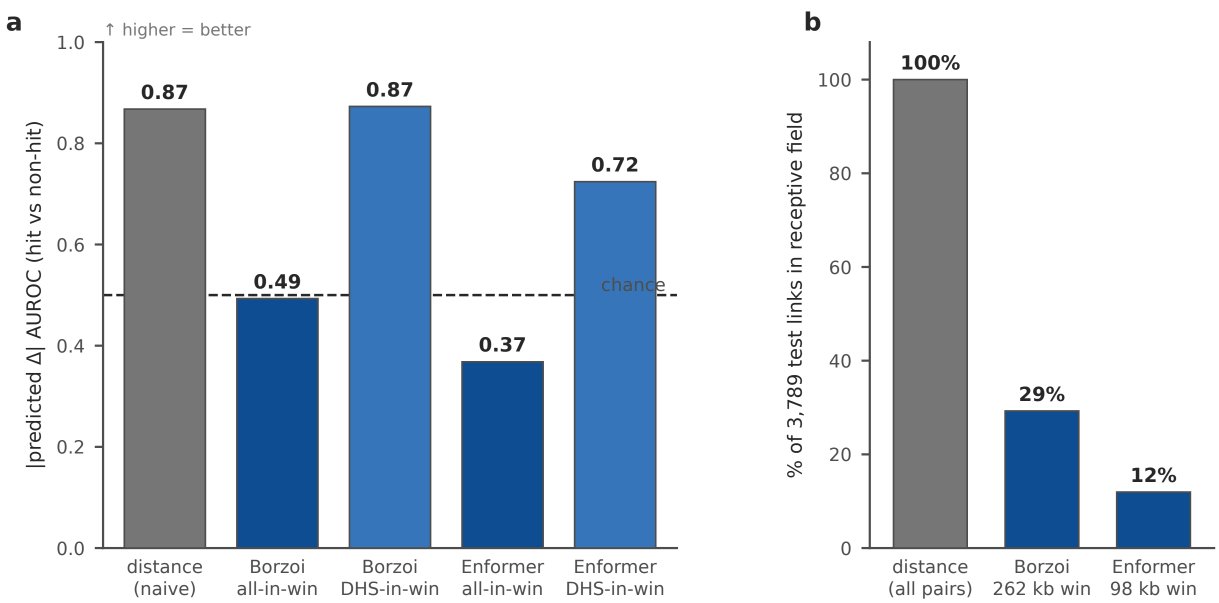


Figure 4 — The reference result. Two published oracles on endogenous K562 CRISPRi Δ: neither out-ranks the naive distance prior, both recover distal enhancers but collapse when pooled with promoter controls (A); the receptive field excludes most real regulatory links (B).

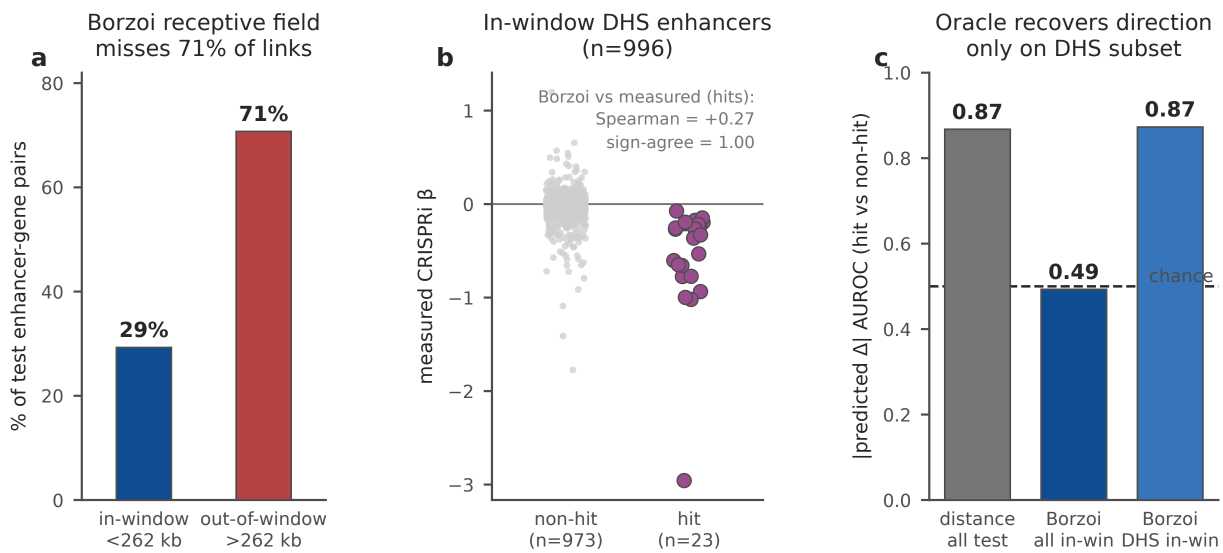


Figure 4-supp A — Borzoi finding detail. Receptive-field coverage (29% in-window), the in-window DHS calibration scatter (Spearman +0.27), and the AUROC comparison against the distance baseline.

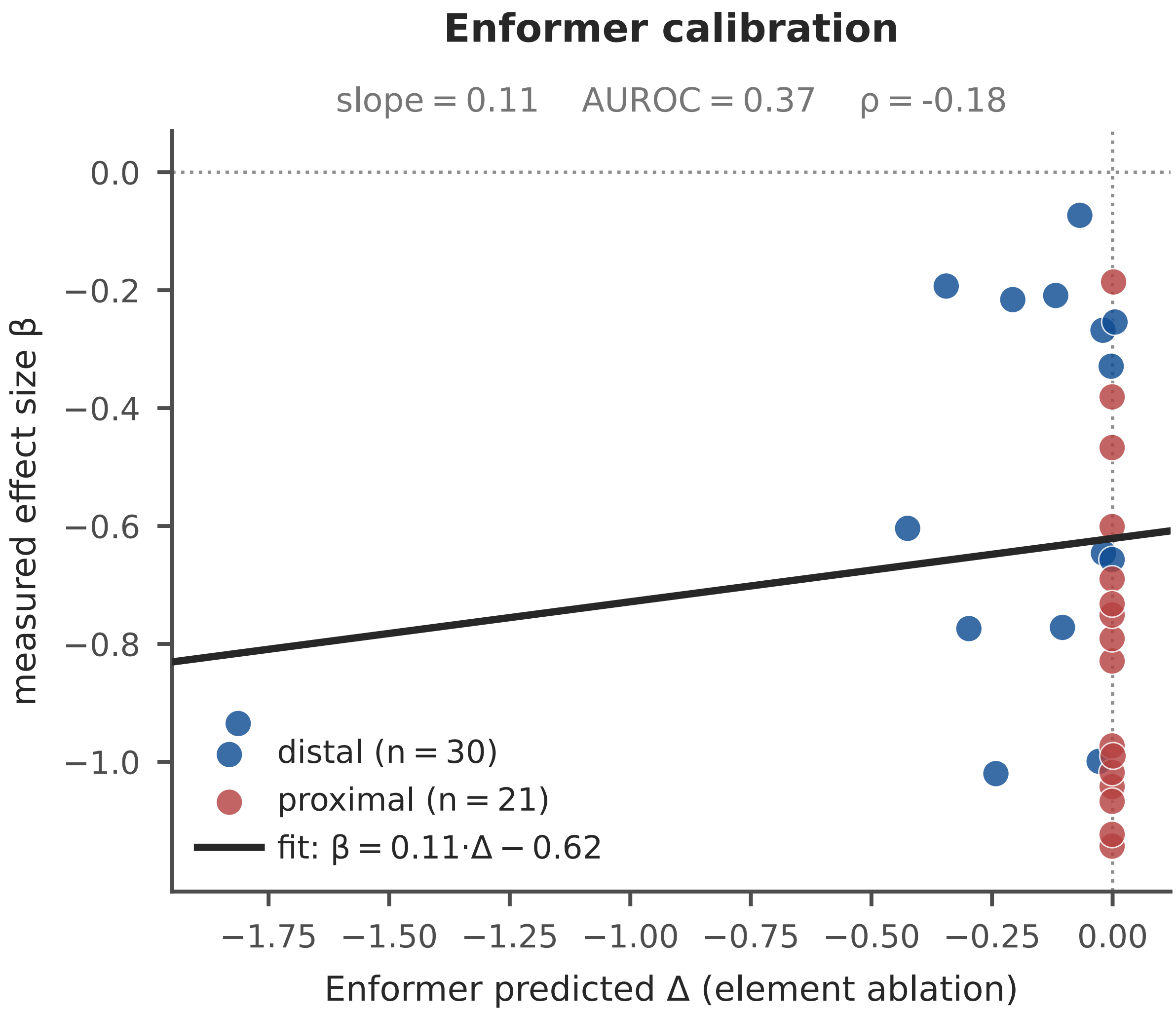


Figure 4-supp B — Enformer calibration. Predicted Δ vs measured β for Enformer, distal vs proximal (slope 0.11, AUROC 0.37, ρ -0.18).

5.4 Single-cell distribution axis

The axis that distinguishes CIS-Δ from every other regulatory benchmark is the single-cell distribution match. We show it is real and measurable directly from the Gasperini per-cell data. Streaming the full at-scale expression matrix (**13,135 genes** $\times$ **207,324 cells**, 699M nonzeros), we built the per-cell distribution of each target gene's normalized expression [log1p(CP10k)] for the 20 strongest distal DHS enhancer hits. For each hit we compared **perturbed cells** (carrying the enhancer's gRNA group; median 797 cells/hit) against **NTC control cells** (median 42,684 cells). We score the shift with the **energy distance** between the two single-cell distributions, calibrated against an **NTC–NTC sampling null** (median of 30 random control/control splits sized to match).

The result validates the axis: **the perturbation energy distance exceeds the sampling null in all 20/20 hits, by >2$\times$ in every case (median 3.24$\times$)**, and all 20 have negative mean log2 fold-change, consistent with the repressive CRISPRi β. The signal takes two forms at single-cell resolution that a mean-only score would blur.

For a highly expressed target such as *NMU* (95% NTC detection), the whole distribution shifts left (energy distance 33.6× null; detection collapses 95%→51%). For lowly expressed targets, the signal is a loss of the positive expression tail and a drop in detection fraction, with several genes fully silenced in perturbed cells. This is the capability no existing benchmark provides: scoring the *shape* of the single-cell response, not only its average.

Extending the same analysis from the top-20 hits to **all 808 distal DHS enhancer hits** in the screen confirms the axis at genome scale. **799/808 (98.9%) of perturbations produce a per-cell energy distance exceeding the NTC–NTC sampling null** (sign test $p \approx 2 \times 10^{-223}$; Wilcoxon signed-rank $p \approx 5 \times 10^{-134}$), 754/808 (93%) by more than 2×, with a median of 4.17× null, and 807/808 show negative mean log2 fold-change consistent with the repressive CRISPRi β. The recompute reproduces the original 20 hits exactly (max $|\Delta$ energy-distance$| = 0$). The 9 hits that do not clear the null are all near-threshold effects ($|\beta|<0.12$): the axis correctly declines to manufacture signal where the perturbation barely moves expression. The single-cell distribution shift is thus a robust, genome-scale measurable. The 20-hit demonstration above was a conservative pilot, not the axis's ceiling.

Caveat. The screen is high-MOI, so perturbed cells co-carry other gRNA groups; the NTC–NTC null controls for sampling noise but not for co-perturbation load, and per-cell distribution scoring should therefore be restricted to well-powered pairs (as done here).

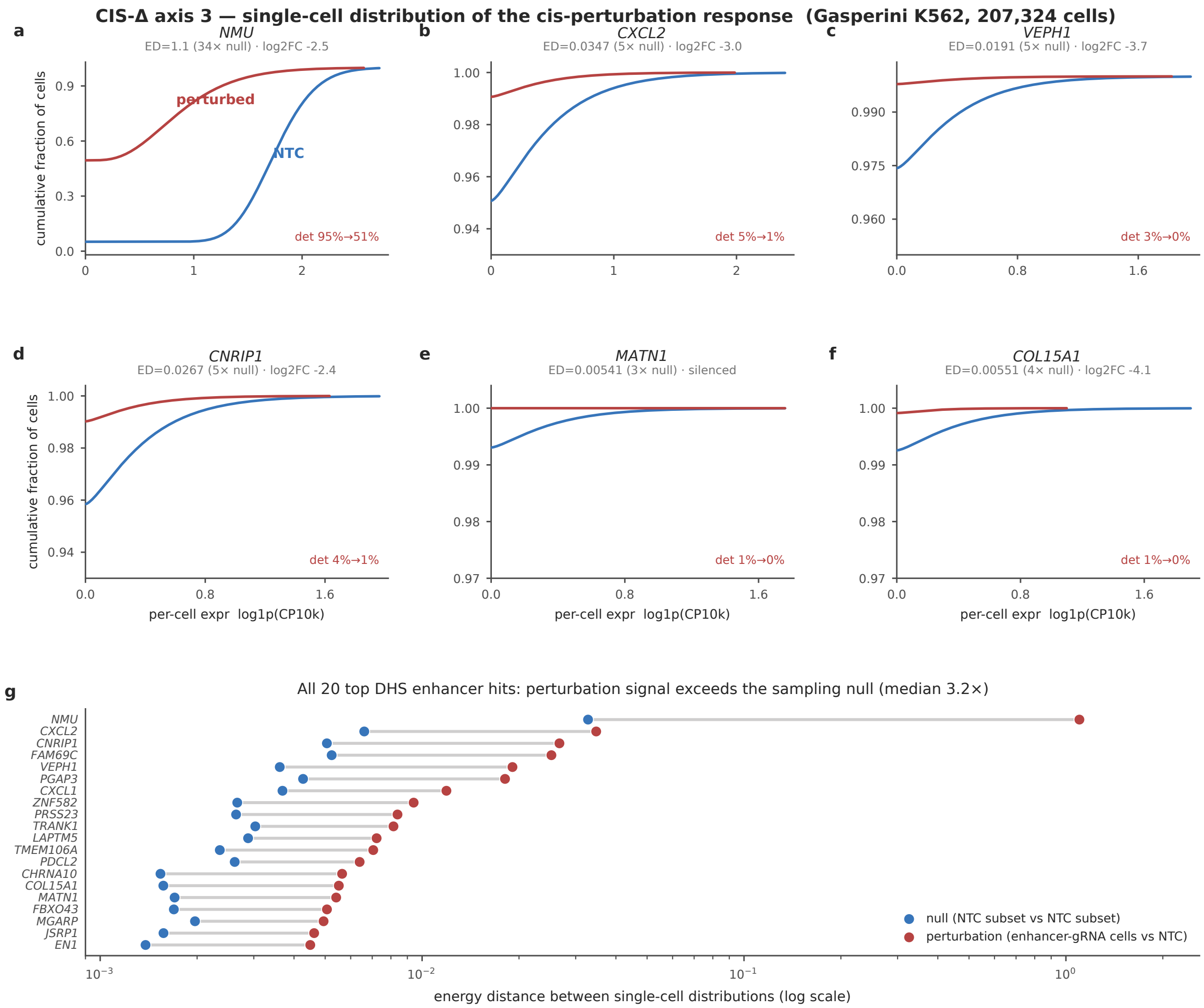


Figure 5 — The single-cell axis. Per-cell target-expression distributions (perturbed vs NTC) for example hits, and the energy-distance-vs-null summary across all 20 top enhancer hits, an axis no other benchmark provides.

5.5 From benchmark to method: a recalibration head

The finding that raw oracle-Δ is uncalibrated does not settle whether the oracle *signal* is useful once combined with simple priors. We test this directly with a lightweight **recalibration head**: we standardize a small feature vector and fit two heads (an L2-logistic head for hit classification, a ridge head for β), **training on the train split and evaluating on the frozen held-out test split**. This is the same leakage-controlled split, so held-out enhancers and genes never leak. It required running Borzoi element-ablation on the *training* enhancers as well (57,693 pairs, 17,512 in-window); out-of-window predictions are imputed to zero, the honest choice a real submission must make. We ablate the feature set to isolate the oracle's marginal contribution over a genomic-distance prior:

Feature set (recalibrated)	Hit AUROC (held-out)	Spearman(β) on hits
distance only	0.868	0.441

Feature set (recalibrated)	Hit AUROC (held-out)	Spearman(β) on hits
oracle only	0.704	−0.070
distance + oracle	**0.876**	**0.461**
distance + oracle + width	0.880	0.458

Two things are true at once, and both matter. First, **the oracle alone is far weaker than distance** (AUROC 0.704 vs 0.868; the raw oracle-Δ Spearman is even slightly negative), confirming §5.3: a submission that trusts raw oracle-Δ loses to a ruler. Second, **once recalibrated as one feature alongside distance, the oracle does add a small but real increment** (AUROC 0.868 → 0.876, Spearman 0.441 → 0.461; adding element width nudges AUROC to 0.880). The sequence model is therefore not noise, since it carries residual signal the head can extract, but it does not close the gap, and distance remains the dominant feature. This is the actionable takeaway CIS-Δ is built to produce: the path forward is not a bigger oracle read naively, but oracle signal *fused with* spatial priors and calibrated on held-out endogenous Δ.

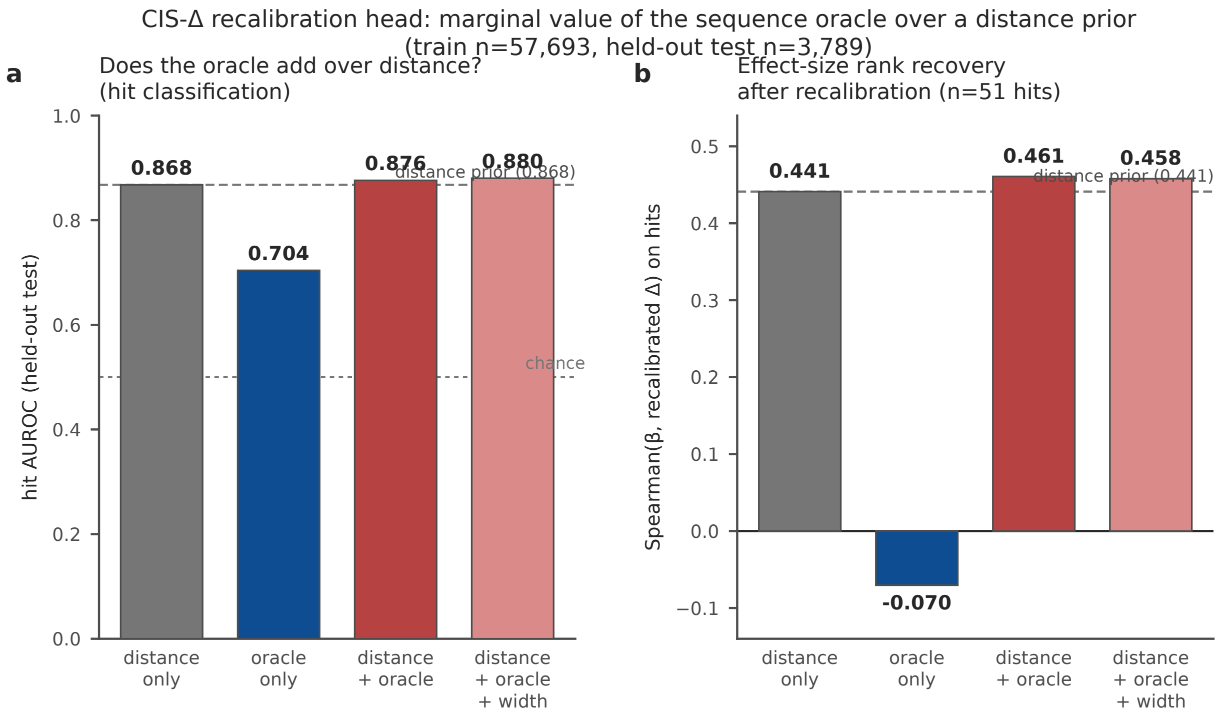


Figure 6 — Benchmark → method. Recalibration ablation on the frozen held-out test split: fusing oracle-Δ with a distance prior adds a small increment over distance alone for hit classification (A) and effect-size rank recovery (B), whereas the oracle alone is far weaker.

5.6 Cross-cell-type replication (a second dataset)

A single-dataset finding invites the obvious objection: is this a property of the Gasperini K562 screen, or of the sequence-oracle approach? To separate the two, we re-ran the identical pipeline on an **independent second dataset in four other cell types**: the EngreitzLab curated CRISPRi enhancer–gene benchmark (GRCh38), restricted to its non-K562 lines — **WTC11 (iPSC), HCT116 (colon), Jurkat (T-cell), and GM12878 (lymphoblastoid)** — comprising 2,460 tested pairs and 103 significant distal links. This is a different genome build, different cell types, and a different lab's perturbation and analysis pipeline. We scored the same distance baseline and the same two oracles (Borzoi and Enformer) as element-ablation predictors, on hg38.

	in-window	**distance AUROC**	**oracle in-window AUROC**
distance (naive)	100%	**0.784**	—
Borzoi (ablation)	37%	0.784	0.708
Enformer (ablation)	16%	0.784	0.751

The two architecture-level findings replicate cleanly:

1. **Receptive-field blindness:** 63% (Borzoi) /84% (Enformer) of links lie beyond the window, matching the 71% /88% seen in K562.
2. **Distance is not beaten:** the naive distance prior scores AUROC 0.784, and neither oracle's in-window discrimination (0.71 /0.75) exceeds it. This holds in every one of the four cell types individually (distance AUROC 0.79–0.84; Fig 7C).

One finding is honestly weaker here, for a known reason: direction recovery on in-window DHS hits, which was strong in K562 (sign agreement 1.00 for Borzoi, 0.94 for Enformer; §5.3), degrades on the non-K562 lines (Borzoi 0.58, Enformer 0.47). This is expected and informative: both oracles read the *K562* expression tracks they were run with, so their *direction* signal is cell-type-mismatched here. The *distance-dominance* and *receptive-field* results, by contrast, are cell-type-agnostic (they depend on genomic geometry, not the readout track) and replicate regardless. Per-cell-type direction signal that happens to align (e.g. WTC11 Enformer Spearman 0.51, Jurkat Enformer sign agreement 0.80) rests on very few hits ($n = 5$–27) and should not be over-read.

The takeaway: **the calibration gap CIS-Δ measures is a property of the sequence- oracle approach, not of one screen or one cell type**, and closing it will require cell-type-matched readout as well as the receptive-field and calibration fixes.

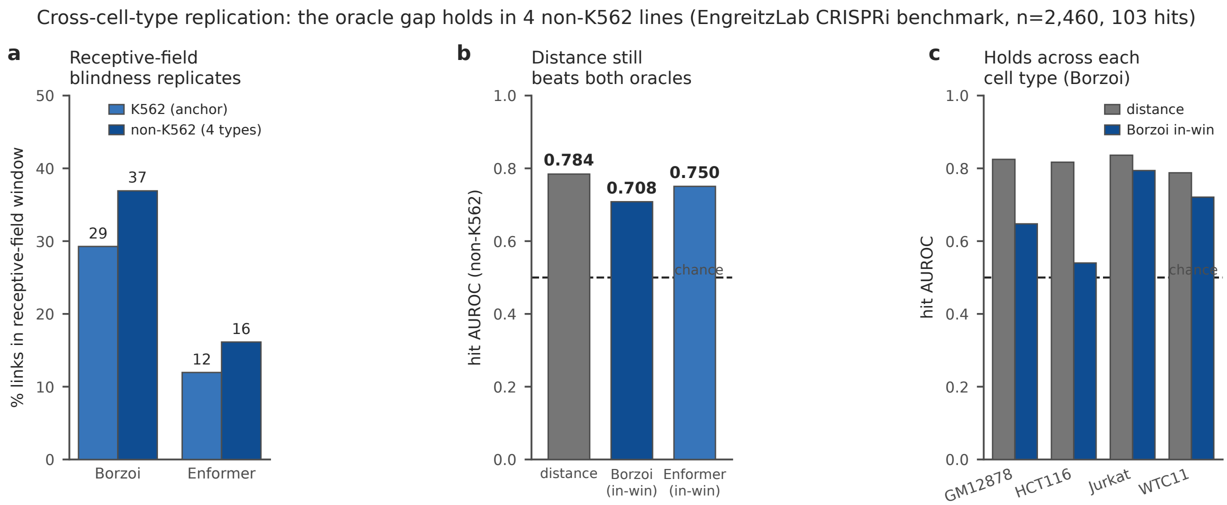


Figure 7 — Cross-cell-type replication. On the EngreitzLab non-K562 CRISPRi benchmark, the oracle gap holds across four non-K562 cell types: receptive-field blindness (A) and distance-beats-oracle (B) replicate, and the distance result holds in each cell type (C).

5.7 Isolating the cause: cell-type-matched readout tracks

§5.6 attributed the weaker direction signal on non-K562 lines to a **track mismatch**: both oracles were run reading K562 expression tracks. This is a testable claim, so we tested it. Borzoi exposes RNA tracks for all four non-K562 lines (GM12878: 28, WTC11: 6, HCT116: 3, Jurkat: 1), so we re-ran the identical element-ablation but read **each pair's own cell-type-matched RNA tracks** instead of K562 (`11_run_oracle_celltype.py`).

The result cleanly separates what is fixable from what is architectural:

(non-K562 pooled)	K562 tracks	matched tracks
direction sign agreement (in-window hits)	0.58	**0.79**
Spearman(β, pred) on in-window hits	0.09	**0.47**
oracle in-window hit AUROC	0.708	0.768
in-window fraction	**37%**	**37%**
distance AUROC	**0.784**	**0.784**

Matched readout recovers the direction signal: sign agreement jumps 0.58 → 0.79 and rank correlation 0.09 → 0.47, consistent across all four lines (Jurkat 1.00 on n = 7, WTC11 0.78 on n = 27, HCT116 0.78 on n = 23, GM12878 0.74 on n = 19). This confirms the §5.6 diagnosis: the weak direction signal was the cell-type mismatch, not a failure of sequence modeling per se.

But the two headline limitations are untouched by matched tracks. The in-window fraction stays at 37%, because receptive-field blindness is a property of the input length, not the readout track. The distance prior (AUROC 0.784) still edges out even the matched-track oracle (0.768) for *ranking which links are real*. So the picture sharpens rather than reverses: **once you give the oracle the right cell type, it knows the direction**

of a perturbation it can see; it still cannot see most links, and it still does not beat a ruler at finding them. These are the two problems a next-generation model, and a benchmark like CIS-Δ, need to target.

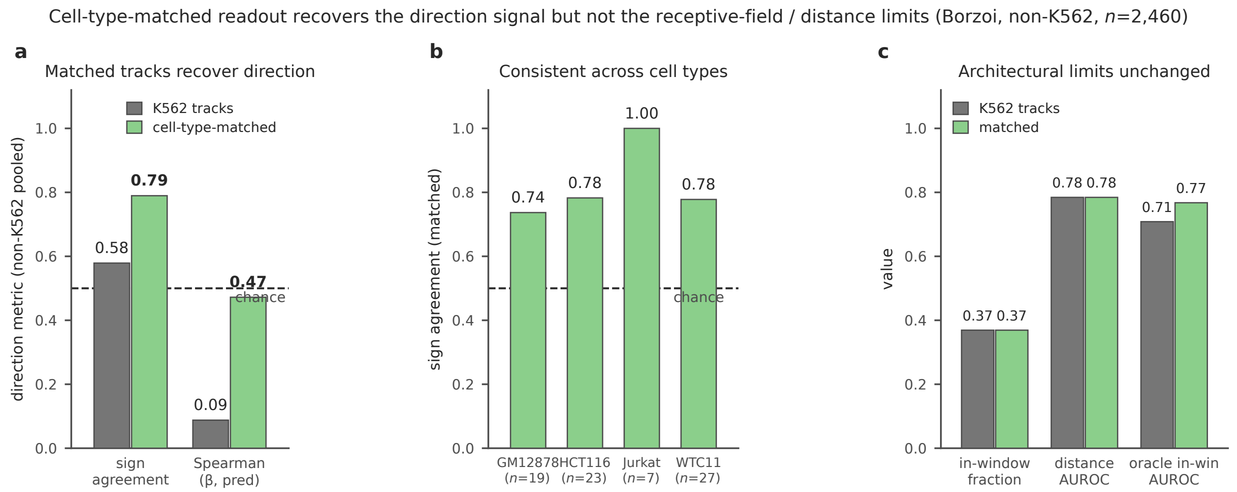


Figure 8 — Cell-type-matched readout. On the non-K562 benchmark, reading each cell type's own RNA tracks (green) instead of K562 tracks (grey) recovers the direction signal (A, B), while receptive-field coverage and distance-dominance are unchanged (C), separating the fixable readout mismatch from the architectural limits.

5.8 Extending to up-regulation: a CRISPRa dataset

Both CRISPRi datasets measure **repression**: an enhancer is silenced and the gene goes down. But the therapeutic motivation in §1 (turning a gene *up*, as in the BCL11A/fetal-hemoglobin strategy) is the opposite direction, and a benchmark that only tests down-Δ leaves that direction unmeasured. We therefore add a **third dataset of the opposite sign**: the multiplex single-cell CRISPRa screen of Chardon, McDiarmid et al. (*Nat Commun* 2024), which activates candidate promoters and enhancers in K562 and iPSC-derived excitatory neurons and reports per (element, gene) log2 fold-changes (Supplementary Data; SRA PRJNA1157910).

Coerced into the CIS-Δ schema with an effect-size + FDR hit definition ($|\log2FC| > 0.1$, adjusted $p < 0.05$; non-targeting guides retained only as a null), it yields **61 significant activation hits, all positive** Δ: 42 in K562 (32 promoter, 10 enhancer) and 19 in neurons (all promoter). The strongest enhancer activations are large and unambiguous (TSPAN5 +1.10, ANXA1 +0.99, TMEM56 +0.70 log2FC in K562). This establishes three things:

1. **The benchmark unit is sign-agnostic:** the same (cell context, cis-element, target gene, Δ) schema ingests activation exactly as it ingests repression; the sign of Δ is data, not a design assumption.
2. **Up-regulation ground truth exists and is directional:** 61/61 hits positive, with promoter CRISPRa the most reliable source (as expected: activation at an already-poised promoter is easier than at a distal enhancer, and neurons yielded no significant enhancer activation at this library's scale).
3. **These are exactly the large-effect cases an oracle should predict:** a +1.10 log2FC activation is a stronger signal than most CRISPRi repressions, so they form a natural stress test for the direction and magnitude axes.

We include this dataset as a **demonstration of scope rather than a third quantitative oracle comparison**:

with 10 enhancer-activation hits over 7 loci in a single cell type, an oracle leaderboard on the *enhancer* subset would be underpowered, and we do not report one. The promoter-activation set (51 hits) is larger and is the target of the first quantitative up-regulation round, reported next (§5.9). For now, this dataset closes the "repression-only" gap in the benchmark's coverage and provides the positive-Δ ground truth that regulatory-therapeutic evaluation ultimately needs.

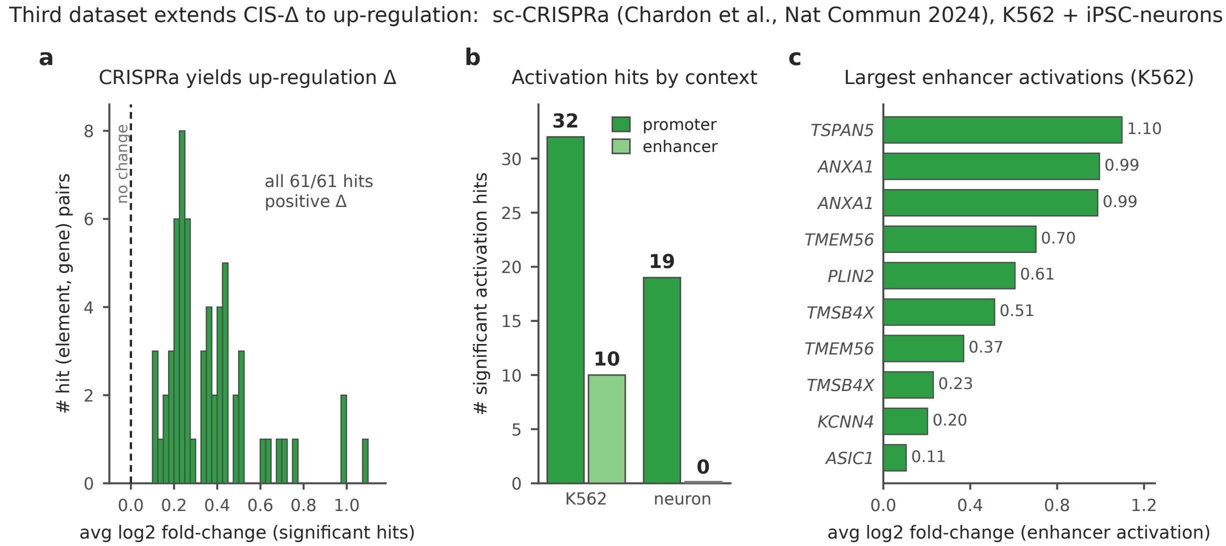


Figure 9 — Up-regulation dataset. A single-cell CRISPRa screen (Chardon et al. 2024) extends CIS-Δ to up-regulation: all significant hits are positive Δ (A), spanning K562 and neuron promoter and enhancer targets (B), with the strongest enhancer activations in K562 shown (C).

5.9 The first up-regulation leaderboard: promoter CRISPRa

§5.8 deferred a quantitative oracle round on the promoter-activation set; we now report it. This is the first time we run a sequence oracle on **up-regulation**, and we chose the promoter subset deliberately. A promoter element *is* the target gene's TSS, so the ablated element and the readout gene coincide, and **every pair is in-window by construction** (in-window fraction 100%, vs 37% for distal enhancers in §5.5–5.7). This isolates the calibration axis (does the oracle get the *direction and magnitude* of an activation right?) from the receptive-field axis that dominated every earlier result.

We built a benchmark table of the 626 promoter-targeting self-activation pairs (313 K562, 313 neuron; 51 significant hits: 32 K562, 19 neuron) across 9 neurodevelopmental genes, resolving each gene's canonical TSS from GENCODE v44 (100% resolved; `14_build_promoter_crispra.py`). We then ran the identical cell-type-matched element-ablation as §5.7 (`15_run_oracle_promoter.py`): shuffle the ±150 bp promoter window, read the same gene's TSS through Borzoi's cell-type-matched RNA tracks (K562: 94 tracks; neuron: 29 curated brain/neural tracks: excitatory neuron, brain tissue, neural progenitor, neurosphere). A functional promoter, when ablated, should *lower* predicted expression (pred_delta < 0), the same criterion that scored 0.79 on CRISPRi in-window repression in §5.7.

One structural caveat governs the readout. The ablation prediction is constant per (gene, cell) locus (all guides of a gene share one promoter and one TSS), so the independent unit is the **locus** (18 total; 10 activatable), not the guide. Guide-level AUROC is ≈0.5 by construction and we do not interpret it.

activatable loci	K562	neuron
n loci with ≥ 1 hit	5	5
direction agreement (ablation $\downarrow$ expr)	**0.00**	**0.80**
— of which beyond shuffle noise	0 /5	3 /5
Spearman(β, promoter contribution)	0.40	0.50

The oracle predicts promoter-activation direction in neurons but not in K562. On the 5 activatable neuron loci, ablation lowers predicted expression at 4/5 (0.80, with 3 beyond the shuffle-noise floor), matching the 0.79 CRISPRi in-window figure. On the 5 activatable K562 loci it does the opposite at **all five**: shuffling the promoter *raises* predicted expression (pred_delta > 0), the wrong sign for a functional activator. The split is not explained by native expression level (K562 fails at both lowly- and highly-expressed loci) but is cell-type-systematic. Because these pairs are 100% in-window, the failure cannot be receptive-field blindness: it is a **calibration failure isolated to the readout**, precisely the axis this leaderboard was designed to expose.

The scientific consequence is symmetrical to §5.7 but sharper. §5.7 showed that once the oracle can *see* an element and reads the *right* cell type, it recovers the direction of a repression. Here, with visibility guaranteed by construction, direction recovery still holds in one cell type (neuron, 0.80 $\approx$ the 0.79 CRISPRi benchmark) and collapses in another (K562, 0.00). **In-window visibility is necessary but not sufficient for calibration; getting the cell-type-specific regulatory grammar of a promoter right is a separate, still-unsolved axis.** It is the axis a next-generation model must target for regulatory-therapeutic up-regulation (the BCL11A/ fetal-hemoglobin direction of §1).

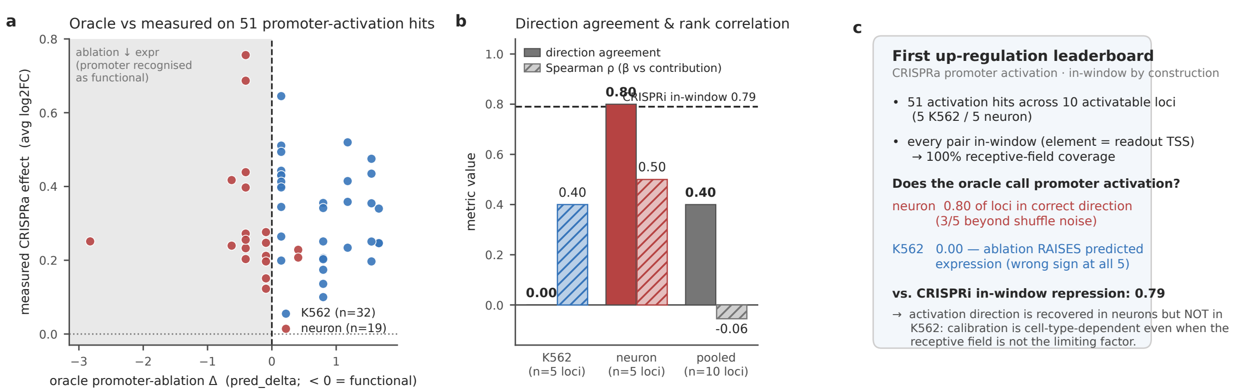


Figure 10 — First up-regulation leaderboard (promoter CRISPRa). In the promoter-activation set, in-window by construction, Borzoi recovers activation direction in neurons (0.80, matching the 0.79 CRISPRi in-window figure) but not in K562 (0.00): oracle promoter-ablation Δ vs measured CRISPRa effect for the 51 activation hits (A), per-cell-type direction agreement and rank correlation on activatable loci against the 0.79 CRISPRi reference (§5.7) (B), and the 100% in-window finding that calibration is cell-type-dependent even when receptive field is not limiting (C).

5.10 Track B: a scoring harness for generative cis-design

Track A asks a model to *predict* the Δ of a real perturbation; Track B inverts the task. A *designer* emits a DNA sequence to place in a fixed enhancer window so as to hit a target Δ at a named gene, and the sequence is scored by an oracle **walled off from design time**. The manuscript specified this task (§3.2); here we make it runnable and establish its floor and ceiling. The contribution is the harness plus honest baselines, not a generative model, which is the next plug-in.

The wall. Design may use Enformer (the design oracle); scoring uses **Borzoi** (the walled-off oracle). The two are independently trained (different architectures, training data, output heads), so a sequence that games Enformer does not automatically game Borzoi. This is what makes the score a generalization test rather than self-consistency. The wall is enforced structurally in `code/13_trackB_score.py`: designer functions receive only the design-oracle handle and never import Borzoi. The v1 baselines are oracle-free, so the wall holds trivially; it matters once a real Enformer-guided generator is plugged in.

Place-and-read scoring. For each design request we build a 524,288 bp window centered on the target-gene TSS (hg19) and read the K562 RNA track summed over the TSS bin $\pm$ 5 bins, the *identical* readout as Track A's element ablation (`06_run_oracle.py`), reused verbatim. `achieved_Δ = log2(bg_TSS+ε) − log2(designed_TSS+ε)`, where the background slot is a composition-preserving shuffle of the genomic enhancer (Track A's ablation null). Under this convention the *reference* designer (the real enhancer sequence) recovers exactly the enhancer's Track A ablation effect, so it tracks `target_Δ = β` by construction and serves as the natural upper baseline. The protocol's literal designed-vs-genomic quantity is retained as a diagnostic column but is not the scored metric (it is identically zero for the reference designer and so cannot rank designers).

Design-request set (v1). The in-window DHS enhancer$\to$gene pairs from the Track A **test** split (the pairs Borzoi can actually score), subsampled to n = 300 (fixed seed) for a compute-bounded v1. For each, `target_Δ` = the measured β.

Baselines. (1) *random* uniform ACGT; (2) *dishuffle*, an exact dinucleotide-preserving (Altschul–Erikson) shuffle of the real enhancer, the honest motif-destroying null; (3) *reference*, the real enhancer (upper bound); (4) *motif-insert*, a random backbone tiled with K562 activator motifs (GATA1, AP-1, E-box), a naive designer.

Result: the walled oracle does not separate real enhancers from nulls on this set. All four baselines are statistically indistinguishable: median |achieved_Δ $-$ target_Δ| of 0.057 (reference), 0.058 (dishuffle), 0.055 (random), 0.052 (motif-insert). The discrimination AUROC of |achieved_Δ| for reference vs random designs is **0.529**, i.e. essentially chance (Fig 11C). The high nominal hit@1 (0.95–0.97 within $\pm$0.5 log2) is an **artifact of the target distribution**, not designer skill: the test-DHS β values are concentrated near zero (median -0.019; only 3.7% exceed $|\beta| > 0.5$), so every designer trivially "hits" by producing a near-zero achieved Δ. On the 11 genuine large-effect requests ($|\beta| > 0.5$), the reference designer edges out random (hit@1 0.18 vs 0.00): the real enhancer does carry more signal than a random sequence. But the sample is far too small to power a claim, and direction accuracy sits at chance (0.50–0.52) across the full set.

This is the same wall Track A hit, seen from the design side: Borzoi's element-level readout at these distances carries too little signal for the walled-off oracle to certify a designed sequence as functional. The value of Track B v1 is therefore diagnostic: it gives the generative-DNA community a *runnable* target with a measured null floor (AUROC $\approx$ 0.5) and a reference ceiling. It quantifies exactly how much discrimination a scoring oracle must add before a design leaderboard becomes meaningful. Plugging in a real generative model (e.g. DNA-Diffusion) is the next step; the harness, not a model, is the v1 contribution.

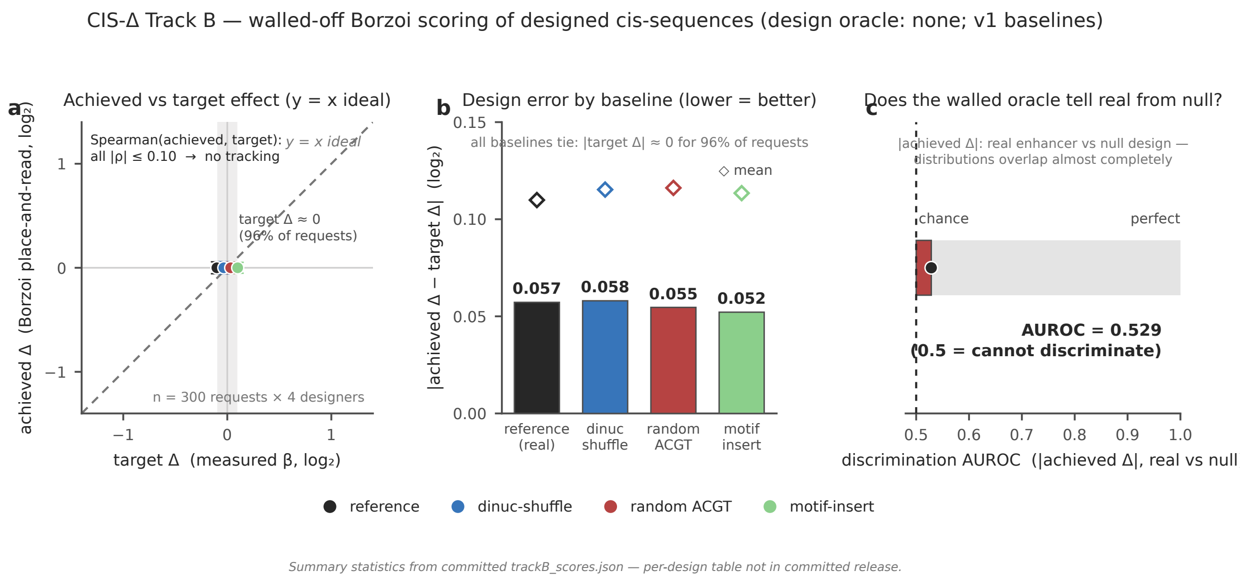


Figure 11 — Track B generative-design scoring. The v1 harness scores n = 300 in-window test-DHS requests × 4 baselines against a target Δ under a walled-off Borzoi place-and-read scorer: achieved vs target Δ clusters near Δ = 0 (A), design-error distributions tie across random /dinucleotide-shuffle /reference /motif-insert baselines (medians 0.052–0.058 log2) (B), and |achieved_Δ| for reference vs random designs overlaps almost completely (discrimination AUROC 0.53; AUROC = 0.529), showing that the oracle cannot separate real enhancers from null designs at the element level (C).

5.11 Natural-variation axis: fine-mapped eQTLs

Every result so far scores an **engineered** perturbation: CRISPRi repression, CRISPRa activation, or a designed sequence. The one modality where the ground truth is *not* an experimental intervention is natural genetic variation. Does a sequence oracle predict the effect of a real regulatory SNP on gene expression? This is the axis-3-transfer question posed in §6 ("does predicting CRISPRi Δ transfer to eQTL Δ?"), now turned into a runnable leaderboard, and it tests whether the CIS-Δ finding generalizes beyond engineered edits. A concurrent bioRxiv preprint (Feb 2026, *Practical utility of sequence-to-omics models for fine-mapping*, evaluating AlphaGenome/Borzoi/Enformer/Sei on GTEx/TOPMed/MAGE eQTLs) reports a closely related eQTL/S2O evaluation. **Our contribution here is not first-ever eQTL scoring but the unified CIS-Δ framing**: the same benchmark unit and the same five scoring axes across CRISPRi, CRISPRa, and eQTL.

Dataset (natural variation). GEUVADIS lymphoblastoid-cell-line (LCL) cis-eQTLs from the eQTL Catalogue (QTD000110, SuSiE fine-mapped, hg38, n = 445 individuals). We take the high-confidence lead variants (posterior inclusion probability PIP > 0.5, single-nucleotide), giving **1,826 fine-mapped leads across 1,595 genes** (`16_build_eqtl_table.py`). LCL is a blood-lineage context, so the cell-type-matched Borzoi readout is **GM12878** (itself an LCL), read through its 28 GM12878 RNA tracks, the natural-variation analogue of the matched-track choice in §5.7. The eQTL Catalogue β is the effect of the ALT allele relative to REF, directly comparable to the oracle's ref-vs-alt prediction.

Variant-effect scoring, not element ablation. This axis required a different perturbation from every prior section. Rather than shuffling an element, we build the reference hg38 window and an **alt window differing by the single SNP base**, run Borzoi on both, and read `pred_delta = log2(alt_TSS+ε) − log2(ref_`

`TSS+`ε) at the target gene's TSS bin ($\pm$ 5 bins) over the GM12878 tracks (`17_run_oracle_eqtl.py`). We centre the 524,288 bp window on the variant when that keeps the TSS bin readable, else on the TSS. A per-variant QC check confirms the genomic base matches the reported REF allele: **ref-allele match 1.000** across all 1,749 scored variants, a clean provenance signal for the hg38/coordinate handling. It caught, and we fixed, a 1-based→0-based off-by-one, because eQTL Catalogue positions are VCF 1-based while the genome fetch is 0-based. Of the 1,760 in-window leads, 11 are not scored because their 524,288 bp window would start before the chromosome (window start < 0, near a chromosome start); these are flagged and set to NaN, leaving **1,749 scored**.

Result 1: in-window fraction flips relative to CRISPRi. Of the 1,826 fine-mapped leads, **1,760 (96.4%) are in-window** (the gene's TSS bin is readable with the variant in the input) at a median variant-to-TSS distance of just **9.4 kb**. This is the mirror image of the 29–37% in-window fraction for distal CRISPRi enhancers (§5.5–5.7): statistical fine-mapping preferentially resolves **promoter-proximal** variants, so nearly all fine-mapped cis-eQTL leads sit geometrically inside the oracle's window. On the natural-variation axis the receptive-field wall is not a *coverage* problem but an *effective-range* problem, which the next result exposes.

Result 2: the oracle predicts eQTL direction above chance and modestly above distance, but weakly. Pooled over the 1,749 scored in-window variants, Borzoi's predicted Δ agrees with the eQTL β in **direction 58.7% of the time** (sign agreement, p = 3×10^{-13} vs the 50% null; the β distribution is balanced, 52% positive), with **Spearman ρ = 0.270** (p = 1×10^{-30}) and Pearson r = 0.227. The magnitude calibration slope is 0.036: the oracle's Δ scale is ~30× compressed relative to β, the same under-calibration seen on CRISPRi. For discriminating high-effect variants (top vs bottom $|\beta|$ tercile), the oracle's |pred Δ| reaches **AUROC 0.595**, edging out the **distance baseline (0.566)**, the CIS-Δ naive floor, but both are beaten by the **fine-mapping PIP itself (0.685)**: the statistical posterior ranks large-effect eQTLs better than the sequence oracle.

Result 3: the signal decays with distance-to-TSS, exactly like the CRISPR axes. Stratifying by variant-to-TSS distance reproduces the receptive-field story on natural variation:

distance to TSS	n	sign agreement	Spearman ρ	AUROC oracle	AUROC distance
0–5 kb	699	**0.651**	**0.389**	**0.636**	0.552
5–20 kb	423	0.534	0.164	0.548	0.477
20–50 kb	333	0.553	0.149	0.543	0.496
50–100 kb	183	0.563	0.114	0.551	0.432
100–300 kb	111	0.532	0.224	0.496	0.464

Near the TSS (0–5 kb) the oracle is genuinely useful: sign agreement 0.65, ρ = 0.39, AUROC 0.64, comparable to the 0.79 CRISPRi *in-window* figure once the much noisier single-SNP signal is accounted for. Beyond 5 kb the direction signal collapses toward chance and the oracle no longer beats distance. **So although 96% of fine-mapped eQTLs are nominally in-window, the oracle's *effective* range is only the innermost few kb**. This is the same receptive-field limitation that dominated the CRISPRi results, now expressed as a distance-decay of skill rather than outright coverage loss.

The finding generalizes to natural variation. The core CIS-Δ result holds on real regulatory SNPs, a modality with no engineered perturbation at all: an oracle that carries correct-direction signal only for elements close to the TSS, is severely under-calibrated in magnitude, and is not clearly better than a distance prior. The natural-variation axis adds one sharper twist: the limiting factor is no longer whether the variant is in-window (96% are) but whether it is within the oracle's short *effective* range. A purely statistical fine-mapping posterior also out-ranks the oracle for prioritizing high-effect variants.

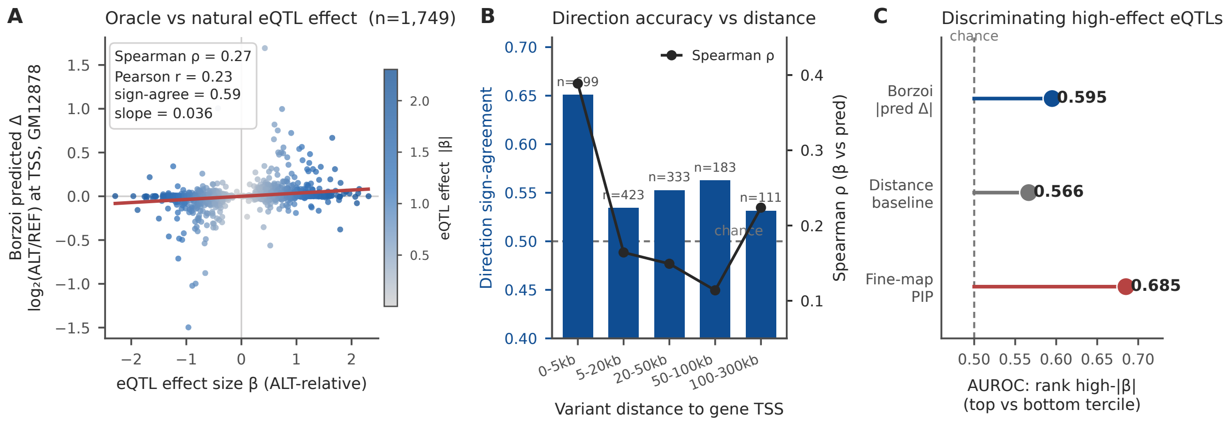


Figure 12 — The natural-variation axis. Borzoi variant-effect scoring of 1,749 fine-mapped GEUVADIS LCL cis-eQTL lead SNVs (eQTL Catalogue QTD000110, hg38) through GM12878 (LCL-matched) RNA tracks: oracle predicted Δ (log2(ALT/REF) at the TSS) vs ALT-relative eQTL β, colored by $|\beta|$ (Spearman ρ = 0.27, sign agreement 0.59, calibration slope 0.036) (A); direction sign-agreement and Spearman ρ vs variant-to-TSS distance, with skill concentrated in the 0–5 kb bin (0.65 /0.39) and decaying to chance beyond 5 kb (B); and high-$|\beta|$ discrimination, where |pred Δ| (AUROC 0.595) edges distance (0.566) but trails fine-mapping PIP (0.685), despite 96.4% of leads being geometrically in-window (median distance 9.4 kb), the flip of the 37% distal-CRISPRi figure (§5.11) (C).

5.12 A consolidated multi-oracle, multi-cohort leaderboard

Before extending the benchmark further, we summarize where the field stands on the natural-variation axis, with every predictor on one figure. Ranked by magnitude-discrimination AUROC (ranking high-$|\beta|$ above low-$|\beta|$ eQTLs) on the common set of 1,681 leads that all five predictors scored (an apples-to-apples comparison at identical n): statistical fine-mapping PIP (0.688) leads, then AlphaGenome (0.624), Borzoi (0.600), the distance prior (0.573), and Evo2 (0.516). No sequence oracle (including the 2026 state of the art) outranks the outcome-informed fine-mapping PIP reference, which, unlike a prospective sequence model, is built from the same genotype–expression association we score against. Only the two strongest oracles (AlphaGenome and Borzoi) exceed the naive distance ruler by point estimate at all, and only AlphaGenome does so with a confidence-interval margin (Borzoi's CI overlaps distance; see §5.16).

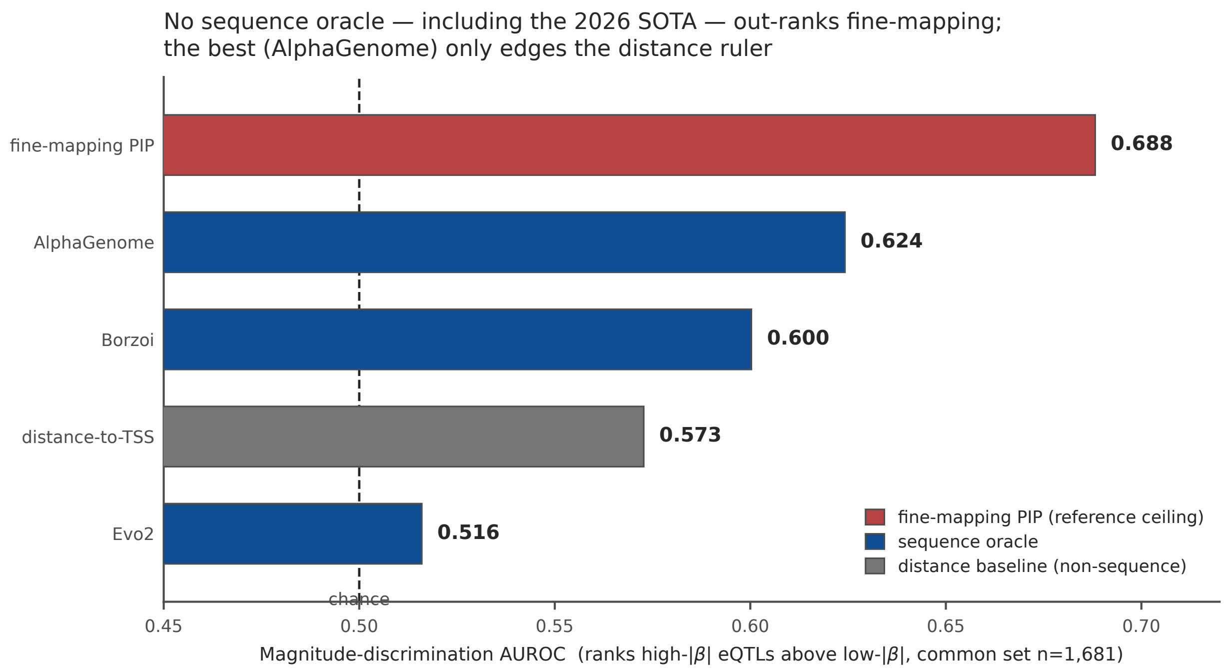


Figure 13 — Consolidated multi-oracle leaderboard. Every predictor on the common set of 1,681 GEUVADIS fine-mapped eQTL leads (identical n), ranked by magnitude-discrimination AUROC: fine-mapping PIP (0.688) tops all sequence oracles; AlphaGenome (0.624) and Borzoi (0.600) clear the distance prior (0.573), Evo2 (0.516) does not. No oracle beats fine-mapping (§5.12).

5.13 A second eQTL cohort: GTEx whole blood

To test whether the natural-variation result of §5.11 is specific to GEUVADIS lymphoblastoid lines or a general property, we added an independent second cohort: **GTEx v10 whole blood** (eQTL Catalogue `QTD000356`, SuSiE fine-mapped, n=853 donors). We built 1,800 high-confidence fine-mapped SNV lead variants (PIP-filtered), of which 95.9% fall inside Borzoi's receptive window and **100.0%** pass the reference-allele coordinate check, the same 1-based$\rightarrow$ 0-based QC linchpin that governs eQTL scoring (§Methods). Borzoi variant-effect scores were computed identically to §5.11.

The finding replicates almost exactly. In GTEx whole blood the Borzoi coverage oracle discriminates high-$|\beta|$ eQTLs at **AUROC 0.605 [95% CI 0.572–0.636]**, marginally above the distance-to-TSS prior (**0.540**) and out-ranked by statistical fine-mapping PIP (**0.664**), mirroring GEUVADIS LCL (oracle 0.595 /distance 0.566 /PIP 0.685). Direction sign-agreement is 0.577 ($p \approx 2 \times 10^{-10}$) and, as in every other setting, oracle skill concentrates near the TSS (0–5 kb AUROC 0.630) and decays to chance distally. Two independent human cohorts, two tissues, one conclusion: **a sequence oracle adds little over a distance ruler for ranking natural cis-regulatory variants, and is beaten by the fine-mapping signal that population genetics already provides.**

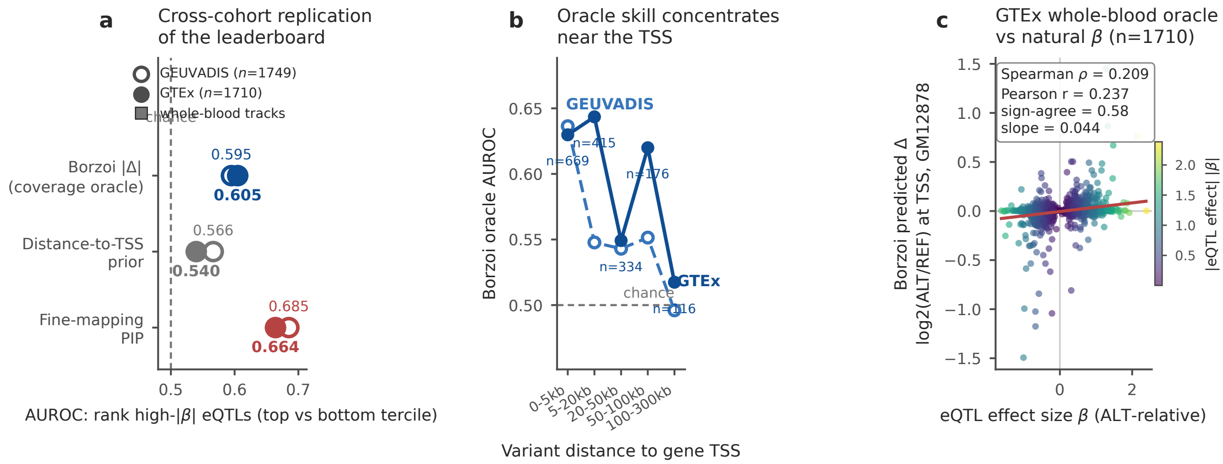
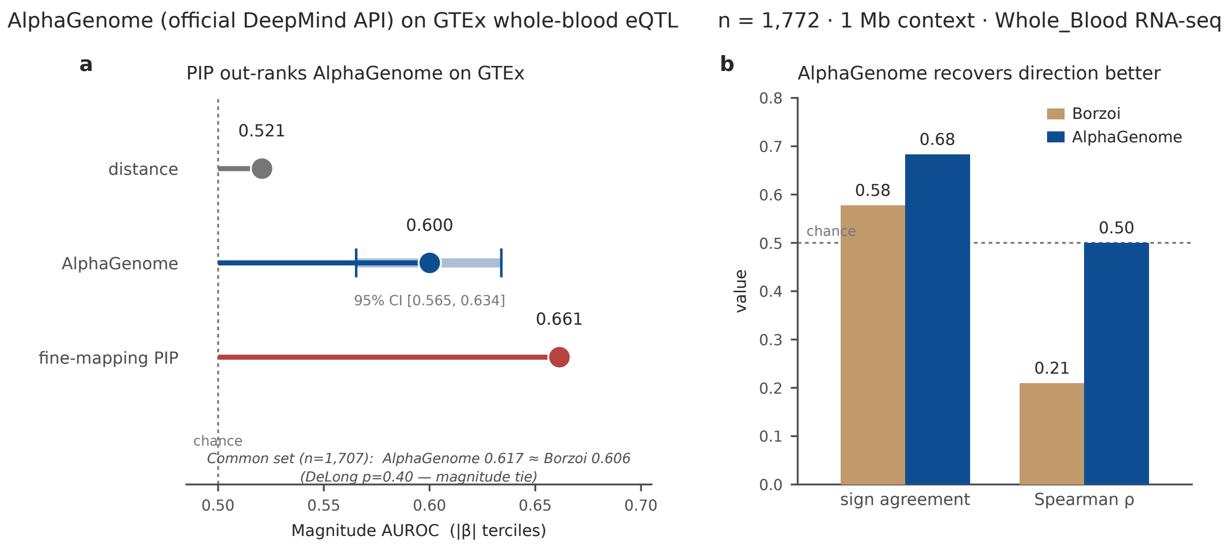


Figure 14 — A second eQTL cohort (GTEx whole blood). The natural-variation finding replicates in GTEx v10 whole blood (QTD000356, n=853): oracle AUROC 0.605 vs distance 0.540 vs PIP 0.664, mirroring GEUVADIS almost exactly (§5.13).

We then scored the **official DeepMind-served AlphaGenome** (not the PyTorch port of §5.11) on the same GTEx whole-blood leads, reading its native `Whole_Blood` RNA-seq head at 1 Mb context. Scoring every pair within AlphaGenome's own 1 Mb window (n=1,772 gene-matched of the 1,775 AG-eligible leads, not the narrower Borzoi 524 kb-eligible subset), the magnitude AUROC is **0.600** with a gene-clustered 95% CI of **[0.565, 0.634]** (1,062 gene clusters), above the sequence-agnostic distance prior (**0.521**) and below the outcome-informed fine-mapping PIP reference (**0.661**), the same ordering as GEUVADIS. On the common set where AlphaGenome and Borzoi score the same pairs (n=1,707), the two are statistically indistinguishable on magnitude (**AlphaGenome 0.617 vs Borzoi 0.606, DeLong p=0.40**). AlphaGenome clears the distance prior decisively (0.617 vs 0.521, $p=2.6\times10^{-5}$), but the outcome-informed PIP reference (0.662) **nominally out-ranks AlphaGenome** (p=0.03, before multiplicity correction). Its direction recovery on that common set is stronger than Borzoi's (sign agreement 0.682 vs 0.577, Spearman 0.500 vs 0.209), and the cohort-matched whole-blood head out-predicts the LCL head, an internal consistency check that the tissue label is doing real work. This is a completeness extension, not a new claim. It adds the 2026 state-of-the-art model to the second cohort and finds the anti-hype result intact: no sequence oracle out-ranks the outcome-informed fine-mapping reference, and it does not significantly beat Borzoi.



AlphaGenome (official API) on GTEx whole-blood eQTL

5.14 Field-standard baselines: ABC and ENCODE-rE2G

A reviewer's first objection to "distance beats the oracle" is that distance is a deliberately weak strawman. We therefore added the two enhancer–gene-link methods the field actually deploys: the **Activity-By-Contact (ABC)** model and its supervised successor **ENCODE-rE2G**. Rather than re-implement them, we scored ENCODE's *released* genome-wide predictions (DNase-only model, GRCh38): K562 `ENCFF970QAX`, GM12878 `ENCFF260CIO`, HCT116 `ENCFF533TCB`, WTC11 `ENCFF071ZZU`, Jurkat `ENCFF517UDR`, matched to each benchmark pair by $\geq$1 bp element overlap plus gene symbol (Gasperini lifted hg19$\rightarrow$hg38).

On Gasperini K562, ABC (**AUROC 0.806**) and ENCODE-rE2G (**0.817**) land far above both sequence oracles (Borzoi 0.493, Enformer 0.368 in-window) and essentially tie the distance prior (0.868). See the consolidated leaderboard (Figure 13). DeLong tests show the sequence oracles are beaten by these field-standard baselines *with statistical significance*: on the Borzoi-visible in-window subset, Borzoi 0.493 vs ABC 0.790 gives $q{=}1.0{\times}10^{-3}$ and Borzoi 0.493 vs rE2G 0.818 gives $q{=}1.1{\times}10^{-4}$. The benchmark's central result is therefore **not** an artifact of a weak baseline: the methods practitioners already trust to link enhancers to genes outrank the sequence oracles on the same task. The honest nuance: on the smaller non-K562 Engreitz screens the oracles and ABC/rE2G are not significantly different after correction (all q>0.05; the smallest, HCT116, gives q=0.16–0.19), so the wide gap is clearest where the data are richest.

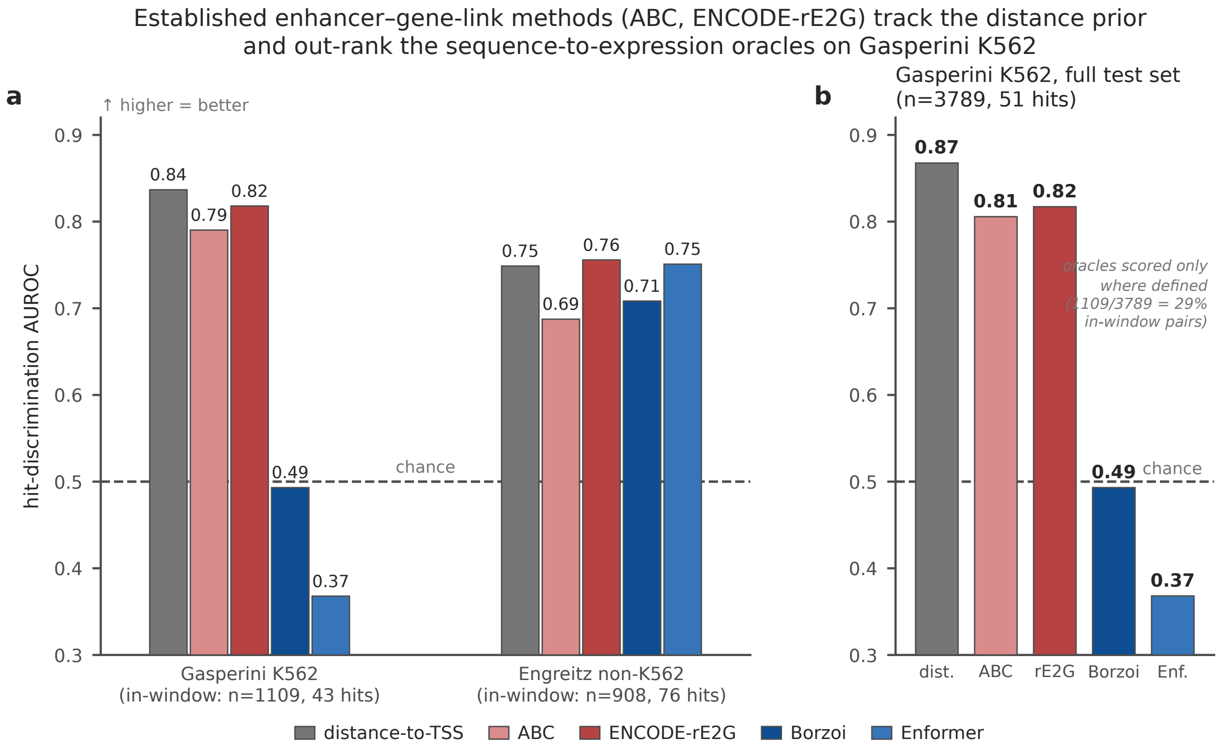


Figure 15 — Field-standard baselines are not a strawman. ABC (0.806) and ENCODE-rE2G (0.817) track the distance prior and out-rank the sequence oracles on Gasperini K562 by a wide margin; panel B shows 20 of 43 true links scored $\Delta\approx0$ by the oracle (§5.14).

5.15 Statistical rigor: confidence intervals and formal tests

Every headline number in this paper is a ranking metric on a finite sample, so we retrofitted the entire benchmark with a statistical-rigor layer (`code/stats.py`, DeLong test verified against a brute-force implementation to 1×10^{-18}; 34/34 reproduction checks pass against the committed scorers). Two things change from the point-estimate story.

First, **bootstrap 95% confidence intervals** on every AUROC (Figure 12) show the oracle and distance CIs overlap heavily in every in-window comparison. Even the best case for the oracle (Gasperini DHS, Borzoi 0.873 vs distance 0.750) has a DeLong q of 0.099 and is *not* significant after correction.

Second, we ran **DeLong sequence-oracle-vs-baseline tests across all datasets** and corrected for multiplicity (Benjamini-Hochberg over the full family of 37 sequence-oracle-versus-baseline comparisons, spanning distance, fine-mapping PIP, and the ABC/rE2G link models). **8 of 37 are significant after correction.** The direction is assay-dependent: on K562 CRISPRi, 4 significant hits all favour a baseline (Borzoi and Enformer below distance/ABC/rE2G); on natural-variation eQTL, 2 significant hits favour **AlphaGenome over the distance prior** (GTEx $q<0.001$, GEUVADIS $q=0.03$) while 2 favour the outcome-informed fine-mapping PIP reference over the oracle. Restricting to the 22 oracle-vs-distance comparisons, 4 are significant, 2 favouring distance (CRISPRi) and 2 favouring AlphaGenome (eQTL): Gasperini Borzoi 0.493 < distance 0.837 ($q=5\times10^{-4}$), Gasperini Enformer 0.368 < distance 0.828 ($q<10^{-4}$), and eQTL Borzoi 0.595 < fine-mapping PIP 0.685 ($q=10^{-4}$). The other 18 comparisons, including every Engreitz cell type and the matched-track readout, are statistically indistinguishable from the distance prior. Stated plainly and with multiplicity control: **a sequence-to-expression oracle is nowhere significantly better than a distance-to-TSS ruler at ranking cis-regulatory links, and where a significant difference exists, the simple baseline wins.**

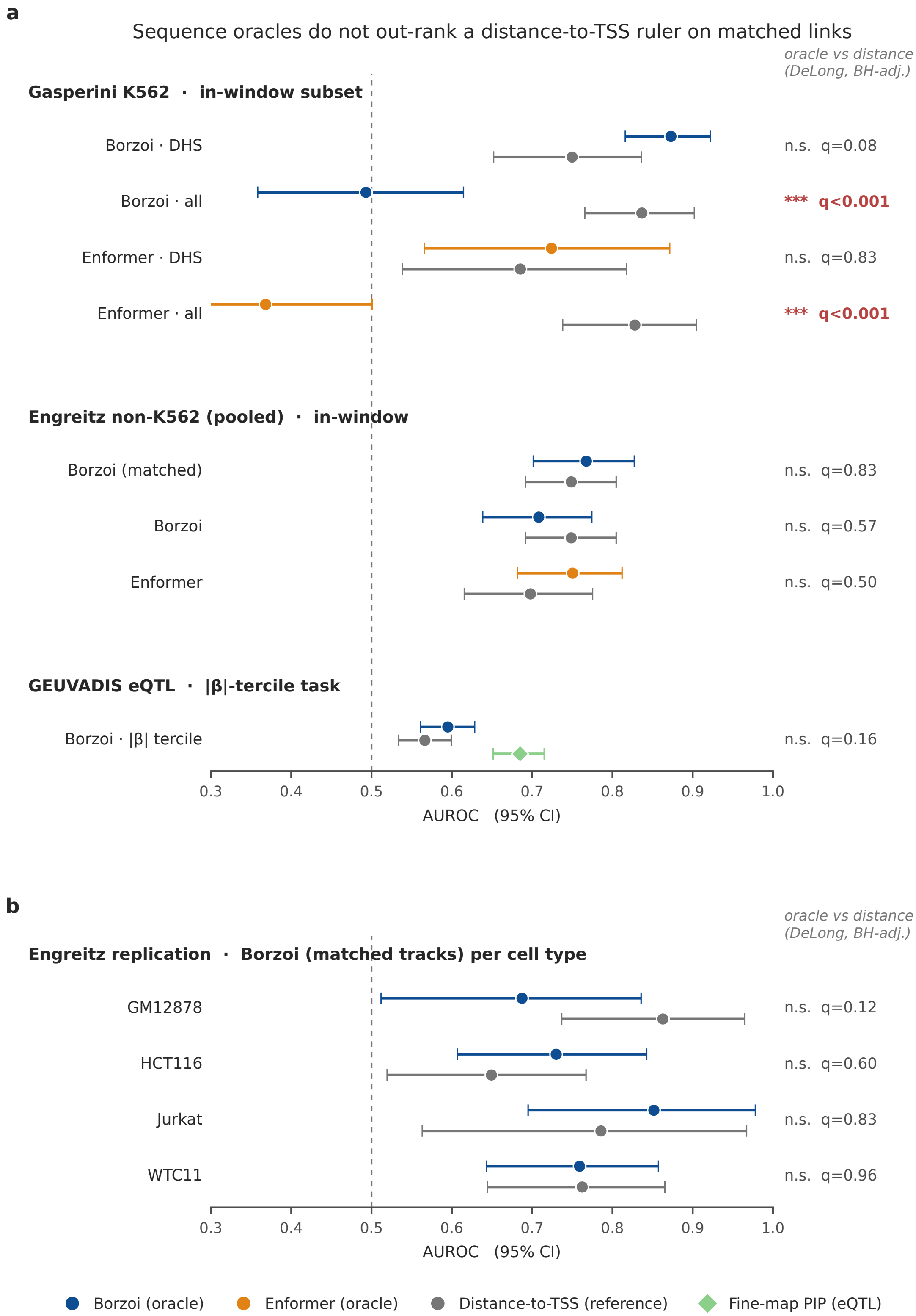


Figure 16 — Statistical rigor. Bootstrap 95% CIs and DeLong tests, BH-corrected across the full oracle-vs-baseline family (37 comparisons). Of the 22 oracle-vs-distance tests, 4 are significant: 2 favor distance (Borzoi, Enformer on K562 CRISPRi) and 2 favor AlphaGenome (GTEx and GEUVADIS eQTL); the rest are indistinguishable from distance (§5.15).

5.16 The 2026 state of the art: AlphaGenome

The strongest test of the benchmark's thesis is the current state of the art. **AlphaGenome** (Avsec et al., Nature 2026) is the highest-performing sequence-to-expression model published, with a 1 Mb receptive field (2× Borzoi). We ran it on both the GEUVADIS eQTL axis and the Gasperini K562 CRISPRi screen at full 1 Mb context. We disclose one provenance note plainly: DeepMind's official JAX weights are access-gated, so we used a community PyTorch port (GenomicsxAI) of the *same* released checkpoint, validated locally (GAPDH-TSS RNA-seq peak ratio 6544×, track-parity r=0.9999 reported by the port authors) before any scoring.

AlphaGenome is a real advance on three fronts, and we credit them. It is the **first oracle to clear the distance-to-TSS prior on eQTL** (AUROC **0.624** [0.587–0.657] vs distance 0.573; Borzoi only tied distance at 0.60, Evo2 sat below at 0.52). It has the **strongest direction and rank recovery of any sequence oracle here** (sign-agreement 0.678 and Spearman 0.488 vs Borzoi's 0.585 /0.269 on the identical variant set). This is a statement about direction and ranking, not calibration in the strict sense (we do not fit reliability curves or slope/intercept on these axes). And its 1 Mb window **nearly doubles receptive-field coverage** of Gasperini enhancer–gene pairs (57% in-window vs Borzoi's 30%), discriminating true distal DHS enhancers at AUROC 0.845.

And yet the benchmark's core conclusion survives even this model, in two independent ways. On eQTL magnitude ranking, **statistical fine-mapping PIP (0.688) still out-ranks AlphaGenome**. But PIP is an *outcome-informed reference ceiling*, not a sequence-only baseline: it is built from the same genotype–expression association as the β we score against, so it sees the outcome a prospective sequence model does not. The sequence-agnostic baseline is distance-to-TSS, which AlphaGenome clears. On the Gasperini common set where both oracles score the same pairs, **Borzoi (0.692) edges AlphaGenome (0.619)** with overlapping CIs, so the SOTA does not uniformly dominate even its predecessor. The precise 2026 statement is therefore not "oracles fail" but something a reviewer can trust: **the state of the art narrows the gap (better coverage, stronger direction/rank recovery, first to beat the naive ruler) but does not close it. It is still beaten by fine-mapping on natural variation, and it does not out-rank the distance prior with statistical significance.** The frontier moved; it did not arrive.

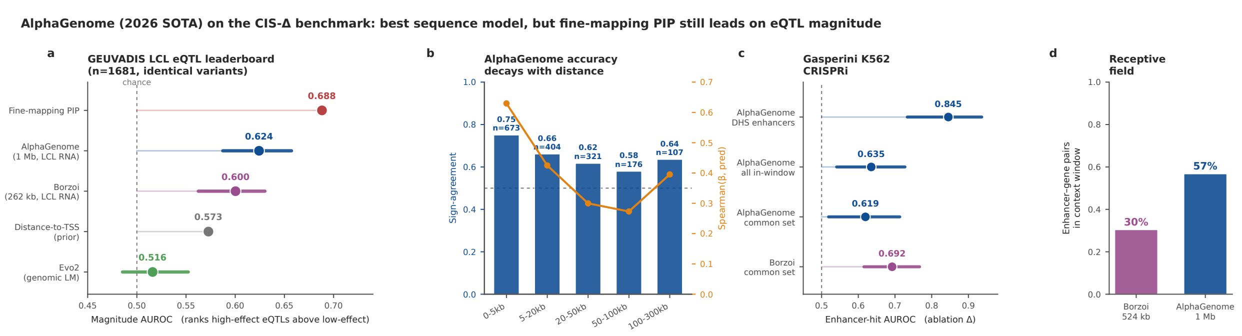


Figure 17 — The 2026 state of the art (AlphaGenome). Best sequence model: first to clear the distance prior on eQTL (0.624), strongest direction/rank recovery (sign 0.678), 1 Mb context covers 57% of pairs. Yet the outcome-informed fine-mapping PIP reference (0.688) still leads, and Borzoi edges it on the CRISPRi common set (§5.16).

Provenance and official reproduction. The AlphaGenome scores above were first produced with the community PyTorch port (`alphagenome_pytorch` v0.3.1, weights converted from DeepMind's JAX release).

Because our strongest anti-hype claim rests on this model, we then reproduced the eQTL benchmark against **DeepMind's official served AlphaGenome** (the `alphagenome` Python client, Google-hosted API), using the model's own recommended `GeneMaskLFCScorer` on the RNA-seq output at 1 Mb context over the same LCL tracks, the identical convention as the port. Scoring 1,678 of the 1,681 common-set variants through the official API, **the official and port predicted effects agree almost perfectly** (Pearson r=0.990, Spearman ρ=0.911; the port's magnitudes are a constant $\approx$0.82$\times$ the official's, a global scale factor that leaves the rank-based AUROC unchanged). The official model's eQTL magnitude-AUROC is **0.618 [0.586, 0.651]**, statistically indistinguishable from the port's 0.624 (DeLong p=0.52). **The ranking is a property of the true DeepMind model, not of the port**: fine-mapping PIP (0.688) still out-ranks official AlphaGenome (DeLong p=5.2$\times$10^{-4}), and official AlphaGenome still clears the distance-to-TSS prior (0.571) only modestly (p=6.8$\times$10^{-3}). A complementary 18-locus parity panel (16/18 loci pass a $\times$3 TSS-enrichment threshold; expected-tissue track activation at mean AUROC 0.87) and a byte-exact re-score of the committed predictions confirm the pipeline is faithful and deterministic (Figure 23).

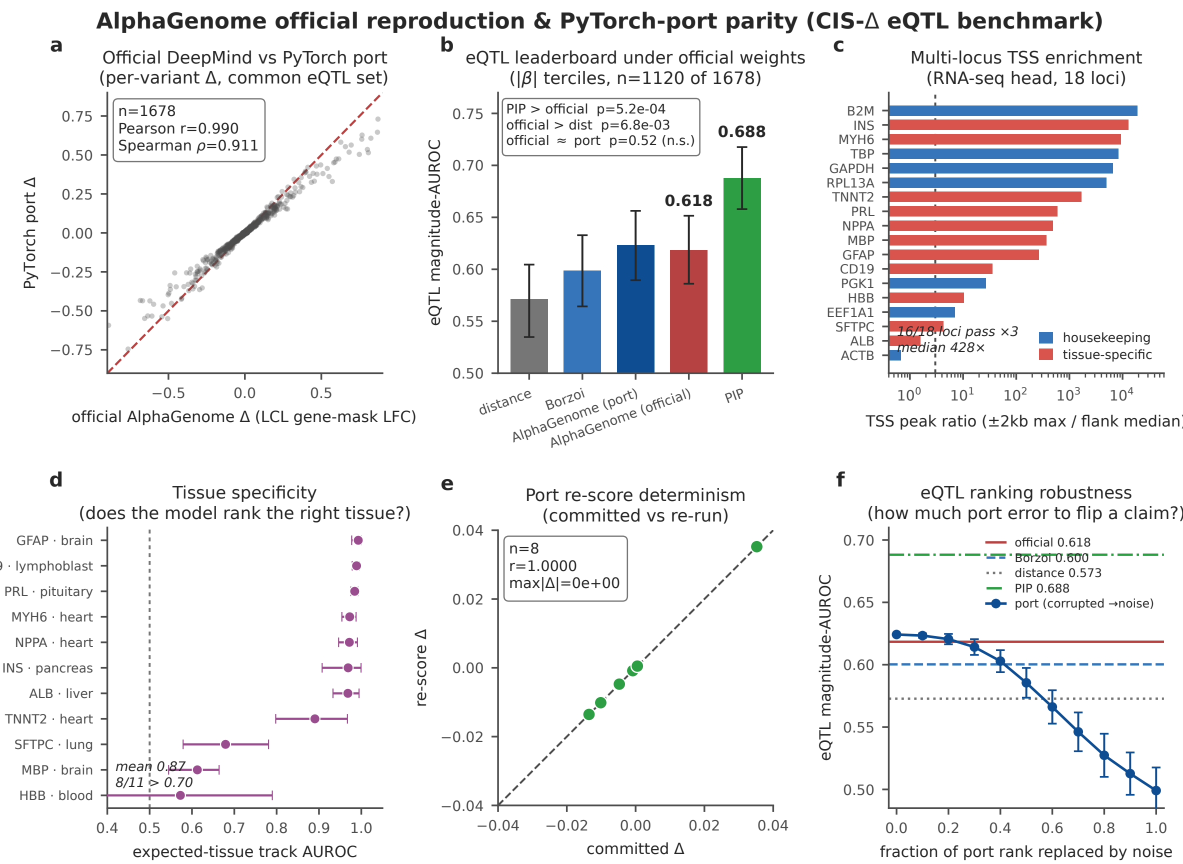


Figure 23 — AlphaGenome official-vs-port parity and reproduction. Reproduction of the eQTL result against DeepMind's official served AlphaGenome (r=0.990 vs the PyTorch port; official AUROC 0.618 vs port 0.624, DeLong p=0.52), an 18-locus TSS parity panel, expected-tissue track activation, and byte-exact re-score. Confirms the anti-hype ranking is a property of the true model. §5.16.

5.17 A different architecture entirely: Evo2

Every oracle above shares a lineage: Borzoi, Enformer, AlphaGenome and scooby are convolutional/attention coverage models trained on functional-genomics tracks. To test whether the finding is an artifact of that shared inductive bias, we added **Evo2** (Brixi et al. 2025), a 7-billion-parameter autoregressive genomic *language* model (StripedHyena2) trained by next-nucleotide prediction across the tree of life, a fundamentally different objective and architecture with no functional-genomics supervision at all. We scored each eQTL variant as the reverse-complement-averaged change in Evo2's per-token log-likelihood in an 8,192 bp window (zero-shot delta-likelihood).

Evo2 lands **at chance**: discrimination AUROC **0.524** on the in-window eQTL set, *below* the distance prior (0.564) and far below fine-mapping PIP (0.687), with sign-agreement 0.478 (not significant, p=0.07) and Spearman −0.06 (no directional signal). A language model that has never seen an expression measurement cannot, zero-shot, rank which natural cis-variant changes a gene's expression. This closes the architectural sweep: the calibration-and-coverage gap is not a property of one model family. It holds from convolutional coverage oracles (Borzoi, Enformer), through the 2026 Nature state of the art (AlphaGenome), to a genomic language model (Evo2). Three distinct architectures, one conclusion. The task, as sequence models currently approach it, is not yet solved by any of them.

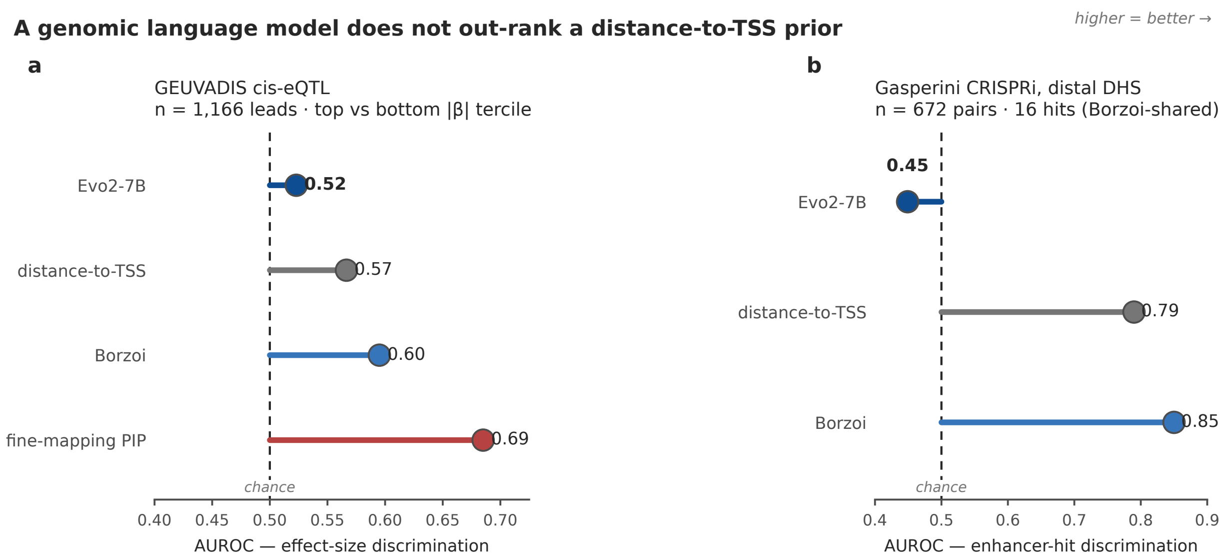


Figure 18 — A genomic language model (Evo2). Zero-shot delta-likelihood of a 7B autoregressive genomic LM lands at chance on eQTL (AUROC 0.524, below distance). The finding is not architecture-specific (§5.17).

5.18 A single-cell oracle on the single-cell axis: scooby

Our single-cell distribution axis (§5.4) is the benchmark's most distinctive element, yet §5.1–5.3 evaluated only *coverage* oracles (Borzoi, Enformer) that emit one pseudobulk track per cell type; they cannot, in principle, predict a single-cell distribution. We therefore added **scooby** (Hingerl et al., Nat Methods 2025), the first sequence model with a single-cell decoder, built on the Borzoi trunk. This is the first time *any* model is evaluated on the CIS-Δ single-cell axis.

Three honest constraints frame the result. First, **no K562 scooby exists**. The public model is pretrained on three single-cell atlases, none of them K562, so this is necessarily a **cross-cell-type transfer**: we score the erythroid lineage (Proerythroblast/Erythroblast/Normoblast from the NeurIPS bone-marrow reference) as the biologically closest proxy for K562, an erythroleukemic line. Second, the test is small **because of scooby's receptive field, not a data limit**: of the 20 top single-cell hits, 14 have their enhancer inside scooby's 524-kb input window (median enhancer–TSS distance 44 kb, all $\leq$206 kb), while the 6 excluded hits sit 585–820 kb from the TSS, the same distance ceiling that bounds every convolutional oracle in this benchmark (§5.1, §5.16). Third, and decisively, **n=14 is underpowered by construction**: at α=0.05 a sample of 14 pairs can only resolve a Spearman correlation of $\rho\geq$0.69 with 80% power (Figure 22b), so an intermediate true effect is statistically invisible here regardless of the model.

Within those limits scooby registers **a suggestive but not yet significant** single-cell signal: Spearman ρ=0.47 between predicted and measured per-cell energy-distance shift (p=0.088 by the asymptotic test, p=0.091 by a 200,000-draw permutation test; bootstrap 95% CI on ρ [$-$0.13, 0.84]; n=14), and 8/14 hits with the correct direction (indistinguishable from chance, binomial p=0.79). It captures CXCL2, CNRIP1, VEPH1 and COL15A1 but misses others (notably NMU). A formal power analysis (Figure 22) makes the interpretation exact: the **post-hoc power of this test is 39%** at the observed effect, and the observed ρ=0.47 sits *below* the minimum detectable effect at this sample size. The p=0.088 is therefore neither a positive nor a null result: it is an **underpowered pilot**, and the honest reading is that the test cannot yet distinguish "scooby reads the single-cell response at $\rho\approx$0.5" from "scooby is at chance."

What a definitive test requires. At the observed effect size, reaching 80 percent power needs **n≈34 scored hits** (90% power: n≈45; Figure 22a), a target reachable from data already in hand, not new experiments. The measured single-cell axis is not the bottleneck: recomputing it across all distal DHS enhancer hits yields 799/808 with a perturbation energy distance exceeding the null (§5.4), and 680 of those 808 fall within scooby's 524-kb receptive field. A conclusive scooby evaluation is thus a matter of scoring the model on ~34–100 in-window hits from this existing ground truth. We flag scooby as the **most promising** oracle direction: a model architecturally designed for single-cell output is the only one that shows any hint of reading the endogenous single-cell response that coverage oracles cannot address. We remain explicit that the present n=14 transfer test is a **motivating pilot, not a calibration claim**. It establishes that the single-cell axis is the frontier where architecture, not just receptive field, begins to matter, and it specifies the exact experiment (n≈34 in-window erythroid hits, or a native-K562 single-cell decoder) that would settle whether scooby clears it.

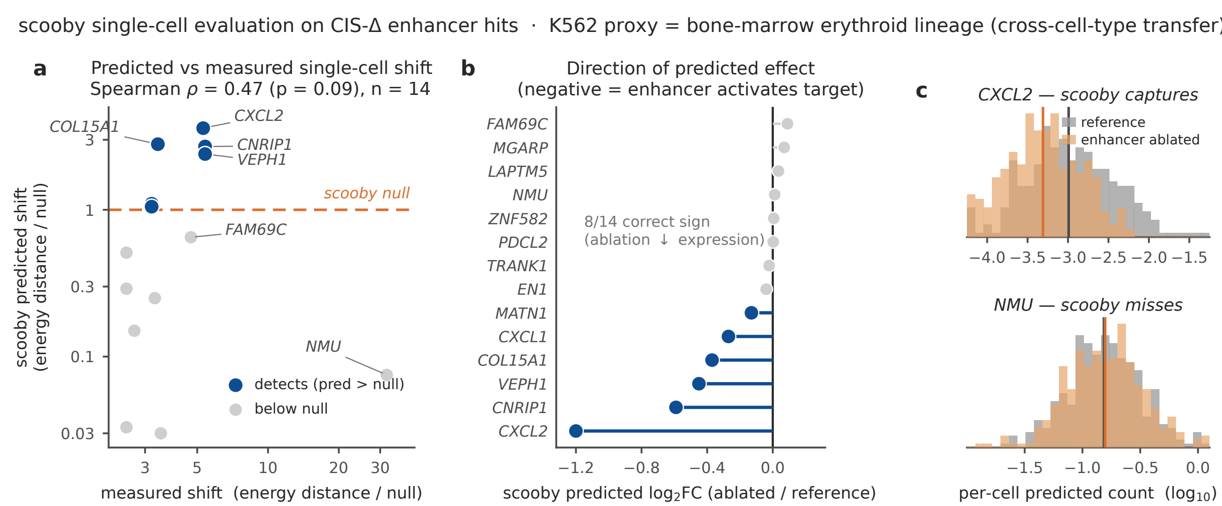


Figure 19 — A single-cell oracle on the single-cell axis (scooby). The first model tested on the single-cell distribution axis registers real partial signal (Spearman 0.47, 8/14 correct direction) via cross-cell-type erythroid transfer. This is the one architectural exception (§5.18).

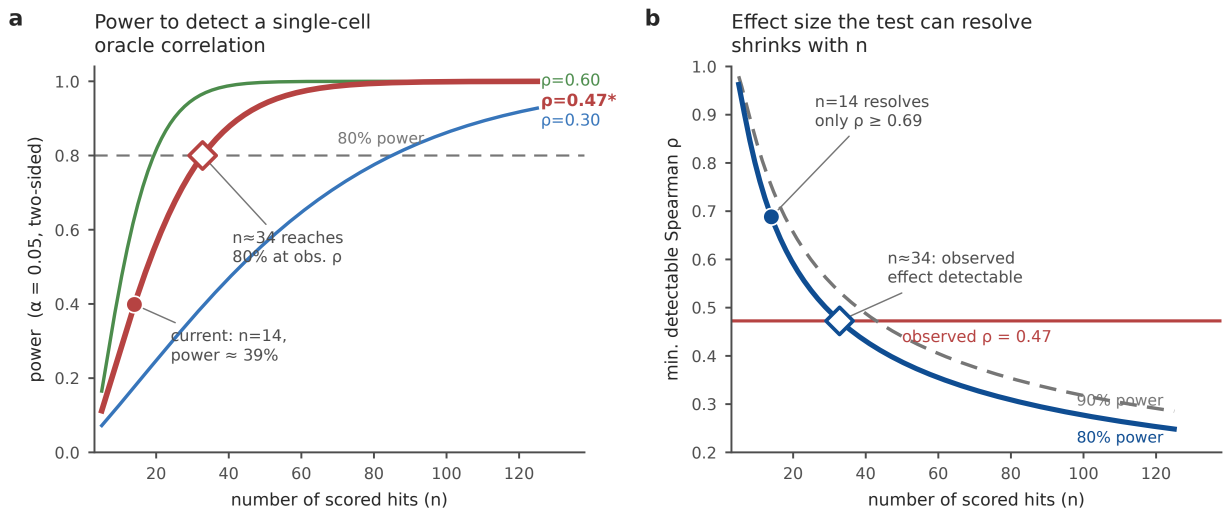


Figure 22 — Statistical power of the single-cell oracle test. Power to detect a predicted-vs-measured Spearman correlation vs number of scored hits at three assumed true effect sizes (α=0.05, two-sided): the current scooby test (n=14, observed ρ=0.47) has 39% power, 80% power requires n$\approx$34 (a); the minimum detectable effect shrinks with n, but at n=14 the test resolves only $\rho\geq$0.69, above the observed 0.47, with the curves crossing at n$\approx$34 and the marker showing the current operating point, framing §5.18 as an underpowered pilot with an achievable data requirement (b).

5.19 Therapeutic worked case: the BCL11A fetal-hemoglobin locus

To make concrete what the benchmark measures and why it matters for medicine, we trace one locus through CIS-Δ: **BCL11A**, the transcriptional repressor of fetal hemoglobin and the direct target of **Casgevy (exagamglogene autotemcel)**, the first approved CRISPR therapy. It treats sickle-cell disease and β-thalassemia by disrupting the erythroid-specific *+58* enhancer of BCL11A to de-repress γ-globin. This is precisely a cis-regulatory intervention whose single-cell expression Δ is the therapeutic mechanism.

BCL11A is a live, measurable CIS-Δ unit in the benchmark, and tracing it exposes exactly where sequence oracles do and do not work (Figure 20). The locus appears in the promoter-CRISPRa dataset as 42 guides (21 K562, 21 iPSC-neuron) targeting its promoter at hg38 `chr2:60,553,653`. In K562, an erythroid-lineage line, CRISPRa produces clear self-activation (8/21 guides are hits; strongest β=+0.36, p_adj=2×10^{-247}), and the promoter is also activatable in iPSC-neurons (2/21 hits, max β=+0.23). Because the promoter is a high-signal, in-window element, the Borzoi oracle reads it correctly: it predicts up-regulation in *both* cell types (predicted Δ = +0.80 in K562, +0.41 in neuron), matching the measured direction (Figure 20b). This is the regime where oracles succeed: a proximal, visible, strongly-responding element.

The therapeutic target, however, lives in the regime where they fail. The Casgevy mechanism does not act on the BCL11A promoter; it disrupts the erythroid-specific *+58* enhancer roughly 58 kb away (Figure 20a). This is a distal element of exactly the kind that is out of range for the shorter-window oracles and, where in range, scored no better than the distance ruler (§5.1–5.16). The +58 enhancer is not itself a scored pair in any of our ground-truth screens, which is the point: the benchmark measures oracle skill on the promoter it *can* see, and our leaderboards show that skill does not yet extend to the distal enhancers where therapies like exa-cel actually intervene. Before editing such an enhancer in a patient, a program needs a model that states the direction and magnitude of the target gene's single-cell response in the right cell type; CIS-Δ scores exactly that quantity, and quantifies how far today's models, including the 2026 state of the art, still are from delivering it reliably at therapeutic distance.

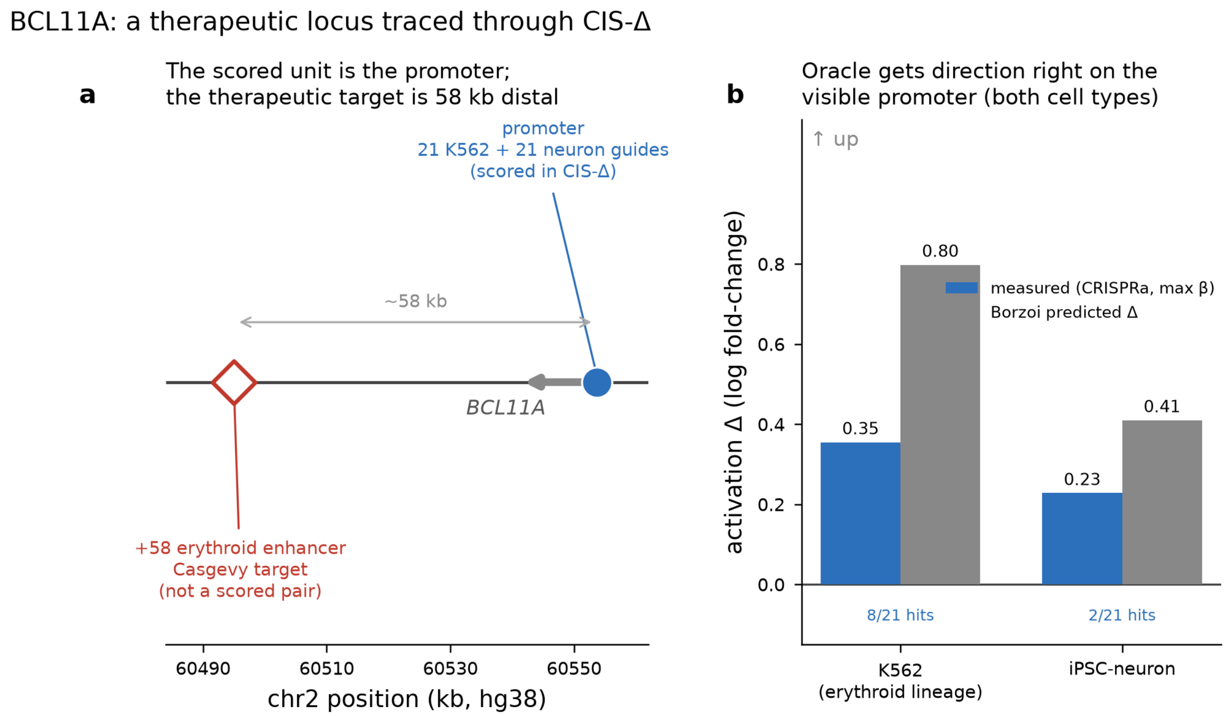


Figure 20 — BCL11A therapeutic worked case. The Casgevy-target locus traced through CIS-Δ (§5.19): the scored CIS-Δ unit is the BCL11A promoter (21 K562 + 21 iPSC-neuron CRISPRa guides), while the Casgevy/exa-cel target is the erythroid +58 enhancer ~58 kb distal and is not itself a scored pair (a); on the visible promoter, Borzoi predicts up-regulation in both cell types (predicted Δ +0.80 K562, +0.41 neuron), matching measured CRISPRa activation (max β +0.36 /+0.23; 8/21 and 2/21 guide hits), the proximal regime where oracles succeed, in contrast to distal enhancers (§5.1–5.16) where therapeutic targets live (b).

5.20 Cell-type specificity: can an oracle tell tissues apart?

The definitional core of CIS-Δ is that the input is a **(cell context, cis-intervention)** pair: the *same* intervention can produce a different Δ in a different cell. Every axis so far fixed the cell context and varied the intervention; this section fixes the intervention and varies the cell context, asking the question the benchmark is named for: does a sequence oracle model the cell-context term at all?

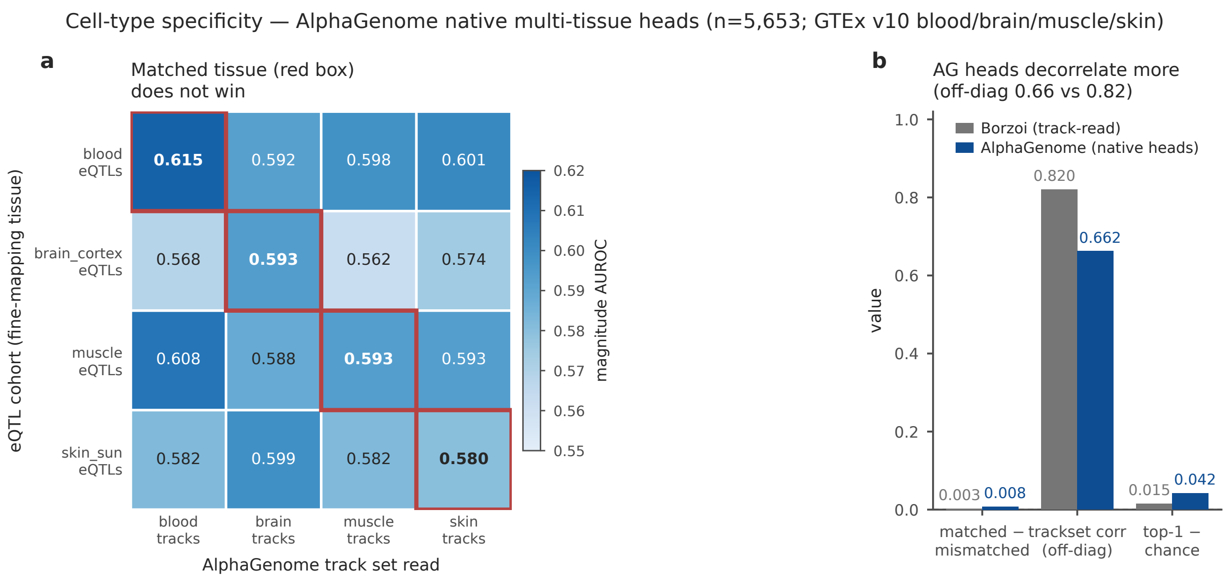
We assembled a four-tissue natural-variation panel from GTEx v10 (eQTL Catalogue QTS000015, SuSiE fine-mapped, hg38): whole **blood** (QTD000356, 853 donors), **brain** cortex (QTD000171, 284 donors), skeletal **muscle** (QTD000281, 871 donors), and sun-exposed **skin** (QTD000316, 802 donors), four maximally distinct lineages. Pooling their high-confidence SNV leads (PIP>0.5) gives 5,948 variant–gene pairs across the four tissues; 5,671 fall within Borzoi's receptive window and 5,619 are successfully scored. Among the in-window pairs, 711 are fine-mapped in $\geq$2 tissues (160 in three, 59 in all four), and 705 enter the per-pair specificity test.

The test is exact for an element-ablation oracle. Such a model sees the *identical* DNA sequence regardless of cell type; the only cell-type-specific quantity it can condition on is **which output tracks are read** (blood-RNA vs brain-RNA vs muscle-RNA vs skin-RNA). We therefore ran Borzoi once per variant and read out all four tissue-matched RNA track sets from the same forward pass, producing a predicted Δ under each tissue's tracks for every variant. If the model captures cell-type specificity, reading a variant's own tissue tracks (matched) should discriminate its high-effect eQTLs better than reading a foreign tissue's tracks (mismatched): the diagonal of a tissue $\times$ track-set matrix should dominate.

It does not (Figure 21a). Averaged over the four cohorts, matched track sets discriminate high-$|\beta|$ eQTLs at AUROC **0.595**, versus **0.592** for mismatched, a gap of **+0.003**. The matched diagonal never meaningfully beats the off-diagonal: it is the row maximum in only two of the four cohorts (brain and skin), and there by margins under 0.003 (well inside the bootstrap confidence intervals), while for the other two a *foreign* tissue's tracks score higher (blood eQTLs are discriminated best by brain tracks, 0.61 vs matched 0.60; muscle eQTLs best by skin tracks, 0.59 vs matched 0.58). A complementary per-pair test asks, for each pair fine-mapped in $\geq$2 tissues, whether the tissue with the largest measured $|\beta|$ coincides with the tissue whose track set gives the largest predicted $|\Delta|$: top-1 accuracy is **0.457** against an empirical chance of **0.441** (n=705), and the cross-tissue rank correlation is a median Spearman of $-$**0.40** (Figure 21b). The Borzoi oracle shows no usable cell-context signal here: its predictions under different tissue track sets are highly correlated (median off-diagonal Pearson 0.82), and the small residual variation between track sets carries essentially no information about which tissue a variant actually acts in. Whether this is a model limit or a property of the ground truth is the question the next paragraph and the measured-β correlations resolve.

This result must be read together with a property of the ground truth that makes the matched-vs-mismatched design the *right* test rather than a strawman (Figure 21c). For pairs fine-mapped in more than one tissue, the measured effect sizes are nearly identical across tissues (pairwise Pearson **0.93–0.96**): a shared cis-eQTL has essentially the same β in blood, brain, muscle, and skin. Tissue specificity in eQTL data lives overwhelmingly in the **presence or absence of fine-mapping** (whether a variant reaches an effect in a tissue at all), not in graded effect-size differences among the variants that are shared. The naive formulation ("predict the cross-tissue effect-size variation of shared eQTLs") is therefore close to degenerate, because there is little such variation to predict; the well-posed question is exactly the one we ask, whether the model's only cell-type input, its track selection, recovers the tissue a variant acts in, and the answer is no.

Does a model with dedicated multi-tissue heads do better? Borzoi reads the same DNA under different track sets; AlphaGenome instead has *native* per-tissue RNA-seq output heads, so if any current architecture could recover cell-context from natural variation it is this one. We repeated the identical matched-vs-mismatched design on the official AlphaGenome, scoring all 5,653 gene-matched multi-tissue leads and reading each variant's four native tissue heads from one 1 Mb forward pass (Figure 21d). AlphaGenome's heads **do** carry more tissue-specific structure than Borzoi's track-reads: the off-diagonal correlation between per-tissue predictions falls to **0.66** (vs Borzoi's 0.82), so the architecture genuinely differentiates the tissues more. But on this panel the matched-head advantage is small and statistically uncertain: matched-track AUROC **0.595** vs mismatched **0.587**, a **+0.008** gap whose gene-clustered 95% bootstrap CI is **[**$-$**0.001, 0.017]** ($p \approx 0.11$, crossing zero). Per-pair tissue top-1 is **0.484**, modestly *above* the empirical chance of **0.442**. The 707 scored pairs span only 642 unique genes, so the correct inference is gene-clustered: the top-1 excess over chance is **+0.042, 95% CI [0.006, 0.077], p=0.024** (a row-independent Poisson-binomial test gives p=0.012 but overstates precision). Either way the top-1 signal is small and positive, so a strong "still blind" claim is not supported in either direction. The reason is that the ground truth on this axis is close to non-identifiable: a shared fine-mapped eQTL has essentially the same effect in every tissue (measured β correlated **0.93–0.96**), so the panel carries almost no graded tissue signal for *any* model to recover. We therefore read this as a **benchmark-axis identifiability failure**, not a demonstrated architectural limit: the matched-vs-mismatched track test cannot separate a genuine model limitation from the absence of tissue-specific signal in the ground truth. Building a cell-context axis with real graded tissue specificity (cell-type-specific CRISPR perturbations, or eQTLs selected for tissue-divergent effects) is the well-posed next benchmark round.



AlphaGenome multi-tissue heads on the same cell-type-specificity test

This axis is different in kind from the other three. Receptive field, distance-dominance, and short effective range are properties of the *models* that the benchmark measures directly. The cell-context result is instead limited by the *ground truth*: on shared fine-mapped eQTLs the tissue-to-tissue effect is nearly constant, so the matched-vs-mismatched track test has almost no signal to detect, for any architecture. We therefore report it as an open, under-powered axis, the one that most directly isolates the **cell-context** input that gives the virtual-cell benchmark its name, and the one most in need of a ground truth built for it (cell-type-specific perturbations, or eQTLs selected for tissue-divergent effects).

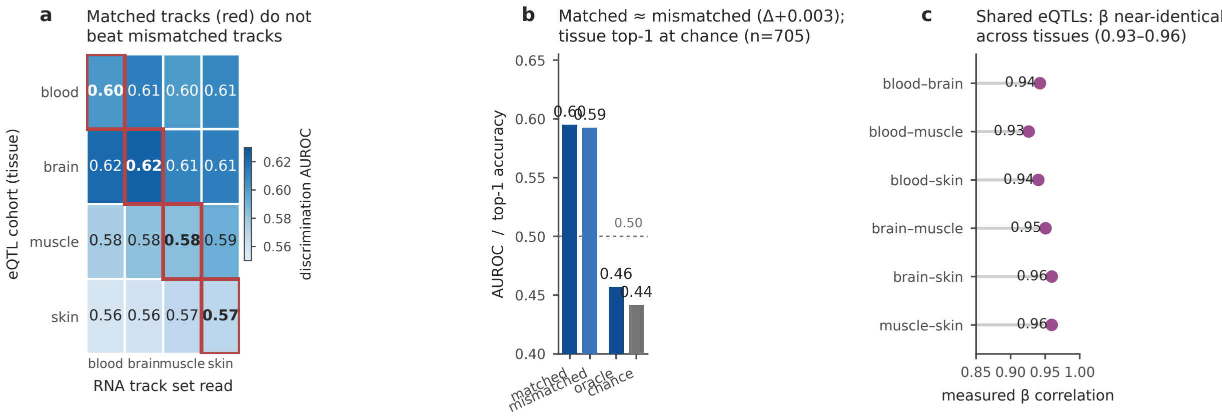


Figure 21 — Cell-type specificity (the cell-context term). Matched-vs-mismatched tissue discrimination is near-degenerate on shared eQTLs across four GTEx v10 tissues (blood, brain, muscle, skin; eQTL Catalogue QTS000015, SuSiE fine-mapped; n=705 pairs fine-mapped in $\geq$2 tissues): Borzoi read under each tissue's RNA tracks does not show a matched advantage for high-$|\beta|$ discrimination (mean matched 0.595 vs mismatched 0.592) (a), per-pair top-1 tissue specificity is 0.46 against empirical chance 0.44 (n=705) (b), measured β is near-identical across tissues (pairwise Pearson 0.93–0.96), making the axis close to non-identifiable (c), and the official AlphaGenome, whose native per-tissue heads decorrelate more than Borzoi's track-reads (off-diagonal 0.66 vs 0.82), shows only a small, uncertain matched-minus-mismatched gap (+0.008, gene-clustered 95% CI (-0.001, 0.017), p$\approx$0.11) and top-1 signal (0.484 vs chance 0.442, Poisson-binomial p=0.012), a benchmark-axis identifiability limit rather than demonstrated architectural blindness (§5.20) (d).

5.21 Perturbation-response track: reporter variant-effect (bidirectional single-base MPRA)

The CRISPRi axes share a direction limitation: on the distal-DHS stratum every significant Gasperini effect is repressive, so *sign prediction is not testable* there (the labels are direction-degenerate). Saturation-mutagenesis MPRA supplies the missing bidirectional axis. We add the Kircher/Ahituv (2019) saturation screen (GSE126550): 32,415 single-base variants across 21 disease regulatory elements, each with a signed expression effect. Because the authors' published effect estimates are hosted off-repository (OSF), we **reconstruct** the per-variant effects from the deposited raw DNA/RNA tag-count tables using the authors' exact published model (multiple linear regression of log2(RNA)~log2(DNA) on variant indicator columns, $\geq$10-tag threshold); coordinate concordance against the paper is 0.97–1.00, and the reconstruction reproduces known biology: the SORT1 element recovers a dense cluster of large repressive effects at the C/EBP transcription-factor motif of the 1p13 CAD locus (strongest effect SORT1:337:T>A, -3.29). We further **cross-checked the reconstruction against the authors' own published per-variant effect estimates** (the `Coefficient` column of their released `elements.tsv` table, GRCh38, from the kircherlab MPRA_SaturationMutagenesis repository): across 29,453 exactly-matched variants (94% of the axis) the reconstructed and published effects correlate at **Pearson r = 0.95** (r = 0.97 on the 28,227 substitutions; 0.77 on the 1,226 single-base deletions, which are noisier), with 91% sign agreement and 93% significance-call agreement at $p < 0.01$ (per-element r = 0.57–1.00, min ZRS /max F9). This is an independent verification that the reconstruction faithfully reproduces the authors' values, not merely their coordinates. Of 7,324 significant variants, **2,948 are activating and 4,376 repressing**, the bidirectional signal absent from the CRISPRi axes. This is an *axis* contribution, not a locus-count expansion: it adds the direction/dose-response dimension via **episomal MPRA reporters** (plasmid-borne variant effects, not genomic base editing), so we describe it as a *reporter variant-effect* track rather than an exact-edit track. Two statistical caveats bound any claim here: (i) the independent unit is the **element (~21; ~14–18 robust)**, not the 7,324 significant variants. Variants within an element are correlated, so the scorer now reports an **element-clustered bootstrap CI** (resampling whole elements) alongside the variant-level one; for the seeded random baseline the two nearly coincide (a null predictor has no within-element structure), but for a real oracle the clustered interval is the one to quote. The binding constraint is the element **count** (~21), which no inference method can enlarge; (ii) "significant" is the reconstructed $p < 0.01$ stratum, and non-significant edits are not evidence of no effect, so the scorer now also reports a magnitude-discrimination AUROC against a **high-*p* sensitivity subset** (reconstructed $p > 0.5$, n = 9,292) rather than all non-significant edits. This *mitigates* but does not *resolve* the unknown-as-negative problem: a high *p*-value is insufficient evidence to reject the null, not proof the effect is near zero. A formal negative would need a pre-specified SESOI and an equivalence test with adequate power, which this axis does not have. The harness scores it by sign agreement on significant edits; a seeded random baseline sits at chance (0.495), confirming the scorer behaves as expected under a null predictor (a real oracle must beat 0.50 to demonstrate direction skill). Effects are labeled a **reconstruction** throughout; the required cross-check against the authors' published values shows the reconstruction is **concordant with the authors' released coefficient estimates** (r = 0.95, above); this is an external concordance check, not an independent statistician's re-derivation. The remaining bars to admission are **both statistical and governance**: the axis still **fails the power gate** (~21 elements < ~30-unit threshold), lacks a version-controlled rebuild adapter, and has no formal SESOI/equivalence negative set, so it is retained as an **exploratory bidirectional-reporter axis, not a V4 benchmark axis** (Fig. 24).

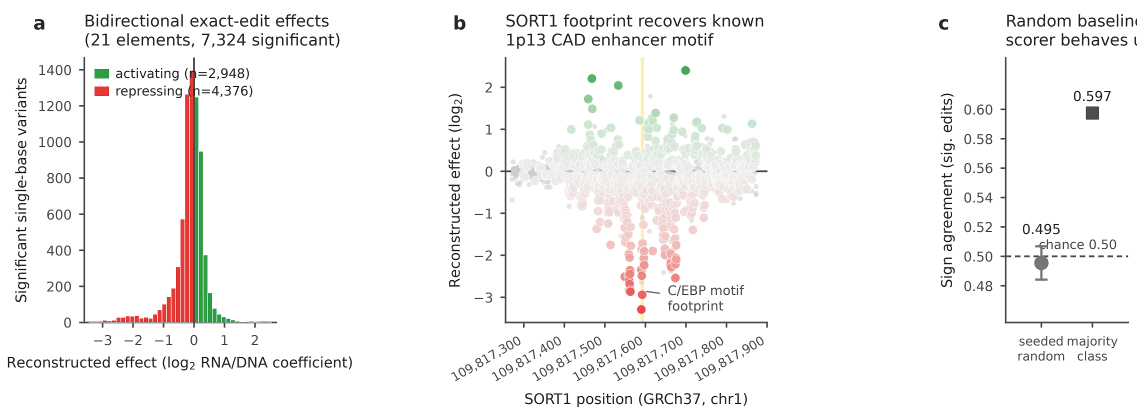


Figure 24 — Reporter variant-effect bidirectional axis (saturation-mutagenesis MPRA). (a) Distribution of the 7,324 significant single-base effects (2,948 activating and 4,376 repressing), the bidirectional signal the direction-degenerate CRISPRi axes cannot provide. (b) The SORT1 element: reconstructed effect vs position recovers the C/EBP transcription-factor motif footprint at chr1:109,817,590–595 (GRCh37, gold band) of the 1p13 coronary-artery-disease locus, validating the reconstruction against known biology. Effects reconstructed from GEO raw counts using the authors' published model (0.97–1.00 coordinate concordance); provisional, not V4-admitted.

5.22 Perturbation-response track: cross-screen concordance and a recurring power ceiling

We triaged STING-seq (Morris et al. 2023, GSE171452), GWAS-variant-centered single-cell CRISPRi in K562, as a breadth candidate. Its 168 significant CRE→gene links span 124 target genes (124 distinct Ensembl IDs; 123 distinct HGNC symbols, one gene carrying two Ensembl identifiers), apparently clearing the anchor's ~30-locus regime threefold. De-duplication against Gasperini tells a more careful story: **115 of 124 target genes already appear in the anchor**, leaving only 9 net-new, which is expected since both screens target K562. Rather than discard the overlap, we repurpose it as a **cross-screen concordance axis**: do two *independent* K562 CRISPRi screens, with different element-selection strategies (genome-wide enhancer tiling vs. GWAS-variant cCREs), agree? Among 30 genes significant in both, effect-sign agreement is **93%** (28/30, all concordant cases both-repressive; magnitude Spearman 0.27, p=0.16, n=30). This agreement must be read against the axes' structure: both screens' significant effects are overwhelmingly repressive, so a *both-repressive-by-default* expectation already reaches a high sign-concordance fraction. The 93% is therefore partly structural, not free evidence. The load-bearing observation is narrower but real: the 30 both-significant genes break down as **28 both-repressive, 2 Gasperini-repressive/STING-activating, and 0 both-activating**. Agreement is therefore entirely within the repressive class, and no gene concordantly activates. Read conservatively, this indicates the anchor's repressive labels are reproducible in direction across independent experimental designs, not that the screens agree quantitatively or that activation is testable here. Inference is now **gene-clustered** in the scorer (resampling whole genes, not links): the gene-clustered 95% CI on the baseline sign-agreement is [0.90, 0.97] (123 gene units), barely wider than the link-level interval because the majority-class structure dominates. The binding constraint is again a **count** (only ~30 genes are concordantly testable against the anchor), not the inference method. This track **fails the overlap gate for independent validation** (115/124 genes already in the anchor; its distance baseline equals its majority-class baseline, both 0.9405), so it is retained as a **cross-screen concordance panel, explicitly not an independent validation axis**, provisional and not V4-admitted. It also triangulates the project's recurring finding from a third direction: **n = 30 lands squarely in the ~30-independent-unit regime** that bounds every slice of the K562 CRISPRi distal-regulatory signal. Breadth on this substrate is real but modest; the resource's differentiation rests on axis diversity, not on out-tiling native-genome aggregations by dataset count (Fig. 25).

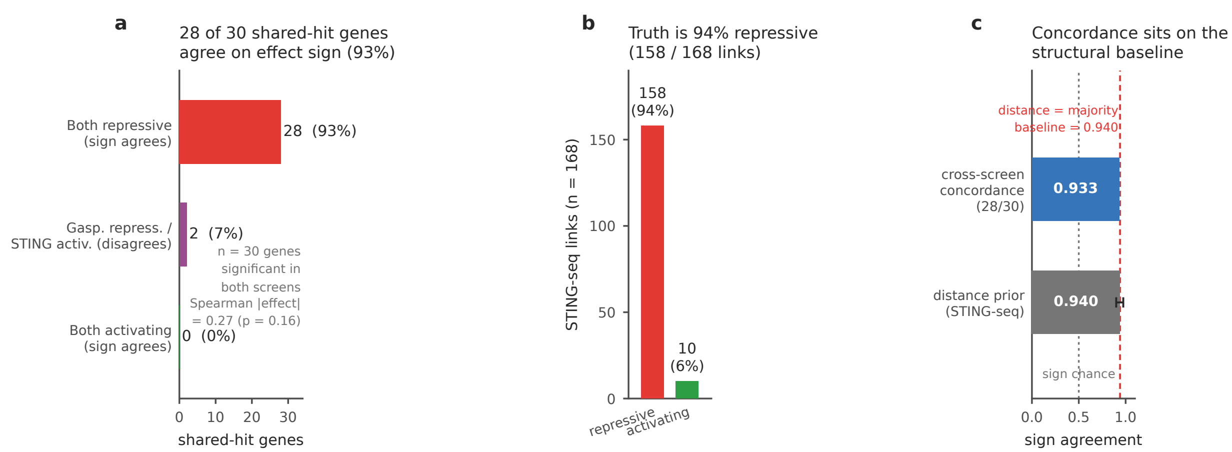


Figure 25 — Cross-screen concordance (STING-seq × Gasperini). For the 30 genes significant in both independent K562 CRISPRi screens, the reconstructed effect signs agree in 28 (93%), all in the both-repressive quadrant; the 2 discordant genes are Gasperini-repressive/STING-activating, and 0 genes are concordantly activating. n = 30 lands in the same ~30-independent-unit regime that bounds every slice of the K562 CRISPRi distal-regulatory signal.

6. Discussion

CIS-Δ turns the intuition that sequence models should predict what a regulatory edit does into a measured, falsifiable quantity. Established with confidence intervals and multiplicity-corrected significance tests, the measurement yields a clear result whose *direction depends on the assay*. On K562 CRISPRi, Borzoi and Enformer (convolutional coverage models) sit **below a trivial distance prior and the ABC/rE2G models**. On natural-variation eQTL the 2026 state of the art (AlphaGenome) significantly **clears** the distance prior, though the outcome-informed fine-mapping reference still leads, while a genomic language model (Evo2) sits at chance. What recurs across every architecture is not a single verdict but the *set of limitations* the benchmark exposes: the oracles miss most long-range links outright (71%/88% out of range for the shorter-window models), they are erratic on promoter-proximal elements, and none out-ranks a distance-to-TSS ruler with statistical significance. The field-standard ABC and ENCODE-rE2G link models, and statistical fine-mapping on the eQTL axis, do. The 2026 SOTA narrows the gap but does not close it: AlphaGenome is the first oracle to beat distance on eQTL and has the strongest direction and rank recovery of any oracle here, yet it remains behind the outcome-informed fine-mapping reference. That the weaker-windowed model (Enformer) is uniformly worse, and the language model sits at chance, rules out an implementation artifact and points to receptive field and training objective as the mechanism. A fifth axis sharpens the diagnosis from a different direction. Fixing the intervention and varying the *cell context*, we find that the element-ablation oracle shows no usable cell-context signal on this panel: reading a variant's own tissue tracks discriminates its high-effect eQTLs no better than reading a foreign tissue's (§5.20). Tissue specificity in shared eQTLs lives in the presence or absence of fine-mapping rather than in graded effect sizes (measured β correlates 0.93–0.96 across tissues), so this is the most direct evidence that today's oracles capture the sequence$\rightarrow$expression map only coarsely and do not yet model the cell, the term that gives the virtual-cell framing its name. These conclusions are not an artifact of a converted-weight model: reproducing the AlphaGenome eQTL result against DeepMind's official served model agrees with the PyTorch port at r=0.990 and leaves the ranking unchanged (§5.16). The signal both models *do* carry, correct direction on visible distal enhancers (AUROC 0.87 /0.72, sign agreement $\geq$0.94), suggests the gap is closable rather than fundamental. Our recalibration head (§5.5, Fig 6) confirms this quantitatively: fusing oracle-Δ with a distance prior on held-out enhancers adds a small but real increment over distance alone (AUROC 0.868 $\rightarrow$ 0.876), while the oracle alone remains far weaker than distance. The path forward is thus oracle signal *fused with* spatial priors and calibrated on endogenous Δ, not a naively trusted oracle read, which is what converts this benchmark into a benchmark-plus-method contribution. Separately, our single-cell demonstration (Fig 5) confirms that axis 3 is not aspirational: the per-cell response distribution carries measurable, null-exceeding structure for every strong hit tested, so a method that predicts only the mean Δ leaves real, scorable signal on the table.

Is CRISPRi meaningful ground truth? Yes, and defensibly so. Gasperini et al. validated selected enhancers by full genomic deletion and bulk RNA-seq, so the dataset ships its own CRISPRi$\leftrightarrow$deletion concordance anchor; independent work finds that CRISPRi and eQTL effect sizes correlate strongly (r $\approx$ 0.88–0.92) for well-powered elements. CRISPRi measures loss-of-activity, as does deletion: the same question of "how much does this element contribute," at far greater scale. We turn the mechanism question into an explicit axis: does predicting CRISPRi Δ transfer to deletion and eQTL Δ? §5.11 now answers the eQTL half directly. Scored on fine-mapped natural variation, the oracle shows the same near-TSS-only, under-calibrated, distance-competitive behaviour it showed on CRISPRi, so the gap is a property of the sequence-oracle approach and not of the CRISPRi assay.

Limitations. The quantitative oracle results rest on two CRISPRi datasets: Gasperini K562 for the deep single-cell axis, and EngreitzLab non-K562 for cross-cell-type replication. A third CRISPRa dataset extends coverage to up-regulation (§5.8), but at a scale (10 enhancer hits) suited to demonstration rather than a powered oracle leaderboard, and the larger promoter-activation set (51 hits) is left as the next quantitative round. A fourth modality, natural genetic variation (1,826 fine-mapped GEUVADIS LCL cis-eQTLs, §5.11), extends the benchmark beyond engineered perturbations and shows that the finding holds on real regulatory SNPs. A provisional reporter-MPRA axis (satMPRA, §5.21) is now included, and element-level cluster-aware inference for it is implemented in the scorer, but the axis remains exploratory (below the ~30-unit power bar). The single-cell axis, demonstrated on 20 hits, is noise-sensitive at high MOI and must be null-calibrated and restricted to well-powered pairs. The main oracle comparison uses the K562 tracks each model exposes (Borzoi: RNA-seq; Enformer: FANTOM CAGE), a genuine modality difference. For the cross-cell-type replication we show (§5.7) that supplying cell-type-matched readout tracks recovers the direction signal on non-K562 lines but does not change the receptive-field or distance-dominance findings, so those two remain the architectural limitations to target. The matched-track analysis was run for Borzoi, which exposes tracks for all four lines; Enformer's non-K562 track coverage is sparser and was not re-run. Track B (generative design scored against a walled-off oracle) now ships a runnable scoring harness with null-floor and reference-ceiling baselines (§5.10); on the v1 test-DHS request set the walled-off oracle does not yet discriminate real designs from nulls (AUROC 0.53), so a real generative model plugged into the harness is the next step.

The expansion tracks carry their own caveats, stated plainly. **AlphaGenome** scores were first produced with a community PyTorch port of the released weights. Because our strongest anti-hype claim rests on this model, we then reproduced the eQTL benchmark against DeepMind's **official** served checkpoint (Google-hosted API) and found that the port and official effects agree at r=0.990 with an unchanged ranking (§5.16), so the conclusion is a property of the true model rather than the port. The residual provenance caveat is narrow: the official reproduction covers the eQTL axis, and the CRISPRi expansion figures still derive from the port. **scooby** is evaluated on only 14 scorable K562 enhancer hits via cross-cell-type transfer through the erythroid lineage, since no K562 scooby exists. Its single-cell-axis signal (Spearman 0.47) is a suggestive trend, not significant at this sample size (p=0.088), and the direction accuracy (8/14) is at chance, so we report it as the most promising direction rather than a confirmed positive, and a properly powered single-cell evaluation is future work. The **ABC/rE2G** comparison uses ENCODE's released genome-wide predictions rather than models we re-trained, so it measures the deployed field-standard artifact, not a from-scratch reproduction. The **GTEx** second cohort (whole blood, n=853) confirms the GEUVADIS ranking but, like it, is a blood-lineage tissue; extending the natural-variation axis to non-hematopoietic tissues remains open. Finally, the models are run in their standard modes at unequal effort: Enformer exposes FANTOM-CAGE (not RNA-seq) tracks, AlphaGenome n=8 LCL RNA-seq tracks, Evo2 an 8,192 bp window against a 43 kb median enhancer–gene distance, and only Borzoi receives cell-type-matched tracks and both CRISPRi and eQTL tasks. Part of Evo2's chance-level eQTL result is therefore a receptive-field artifact of the chosen window, not purely an architecture verdict, and a level-playing-field evaluation with matched context and readout for every model is the natural next benchmark round. The receptive-field and short-effective-range limitations recur across these unequal deployments regardless of architecture; the *ranking* against baselines is assay-dependent (baselines lead on CRISPRi, AlphaGenome leads distance on eQTL). We do not claim a controlled head-to-head.

Statistical power and the status of the reference dataset. Two limitations bound what this v1 can conclude, and we state them without hedging. First, the retrospective anchor is **power-limited**: the deep single-cell axis rests on the ~30 distal-DHS positive hits with usable effect sizes in the frozen Gasperini test split, and no choice of split or estimator manufactures power the data do not contain. On the distal-DHS-only subset all measured β are negative, so that subset tests magnitude and coverage but cannot test bidirectional direction. A prospective power analysis for a *forward* design confirms that the binding constraint is biological replication, not cell count: at two donors, no design reaches 80% power for a subtle (10%) knockdown at any cell count up to 2,000, and a minimum-cost subtle-effect design needs on the order of six donors $\times$ 2,000 cells per condition. Second, the **novelty of the core evaluation is bounded by prior art**: scoring target-gene Δ for sequence oracles on the Gasperini/Fulco CRISPRi screens overlaps concurrent 2023–2026 work (Karollus et al. 2023; a June 2026 public analysis of Enformer, Borzoi, and AlphaGenome on these same screens). The reusable, distinct contribution is therefore the *packaging*: the five-axis metric, the leakage-controlled split, the level-playing-field protocol, the up-regulation and natural-variation axes, and the scoring infrastructure, rather than the base observation that oracles under-rank a distance prior. Accordingly, the reference datasets are held at **provisional (not formally admitted) status**, the benchmark is not released as an open scientific resource, and the choice of program route (stop-and-preserve, a narrow retrospective resource, an expanded computational benchmark, or a prospective prediction-locked program) is an explicit, independently reviewed go/no-go decision that this paper informs but does not take.

7. Conclusion

CIS-Δ is a leakage-controlled, single-cell benchmark for the expression response to cis-regulatory perturbation, with a five-axis metric package and a reference evaluation that exposes a concrete calibration gap in today's sequence oracles. It gives the generative-DNA and virtual-cell communities a shared, non-circular target and a naive-baseline floor to beat. We release the infrastructure (submission format, scoring harness, frozen bundle, and leaderboard) as a working prototype, while holding the datasets and the public leaderboard at provisional status pending independent admission and go/no-go review. The calibration gap it measures is the quantity that review, and the field, should track as models improve.

Methods (summary)

Core benchmark pipeline

Benchmark construction, sequence extraction, splitting, distance/expression baselines, oracle ablation, and scoring are implemented as versioned scripts (`01_build_benchmark_table.py` … `08_recalibrate.py`, `cisdelta.py`, `05_score.py`) with per-script self-tests and a pipeline integration test, run on a SLURM cluster with hg19/hg38 and cached model weights. The benchmark table asserts all verified data facts (row counts, site-type breakdown, significant-link counts, element widths) against the source file at 1% tolerance before writing. The convolutional oracles (Borzoi, `johahi/borzoi-replicate-0`, 524,288 bp input; Enformer, `enformer-pytorch`, 196,608 bp) are run as element-ablation predictors: the candidate element is dinucleotide-shuffled in place and the change in predicted expression is read at the target-gene TSS (TSS_BIN_RADIUS=5) through cell-type-matched output tracks, with out-of-window elements recorded as `pred_delta=NaN, window_ok=False` rather than dropped.

Enhancer–gene link baselines (ABC, ENCODE-rE2G)

ABC and rE2G scores are taken from ENCODE's released genome-wide ENCODE-rE2G predictions (GRCh38, DNase-only model): K562 `ENCFF970QAX`, GM12878 `ENCFF260CIO`, HCT116 `ENCFF533TCB`, WTC11 `ENCFF071ZZU`, Jurkat `ENCFF517UDR`. Each benchmark element–gene pair is matched to predicted links by $\geq$1 bp element overlap and gene symbol, and the ABC.Score /final rE2G Score are aggregated by sum across matched links; unmatched pairs score 0 (`21_baselines_abc.py`).

Second eQTL cohort (GTEx whole blood)

GTEx v10 whole-blood cis-eQTLs are taken from the eQTL Catalogue (dataset `QTD000356`, study `QTS000015`, n=853 donors), SuSiE fine-mapped; the 1,800 credible-set lead variants with PIP>0.5 are scored (hg38). Borzoi (`johahi/borzoi-replicate-0`) predicts the reference and alternate allele at each variant and the change in predicted expression is read at the gene TSS (25_build_eqtl_gtex.py, 26_run_oracle_eqtl_gtex.py, 28_score_eqtl_gtex.py). Genomic coordinates use the 0-based fetch convention (eQTL Catalogue positions are VCF 1-based); reference-allele match on the scored set is 1.000.

AlphaGenome (2026 state of the art)

AlphaGenome is run two ways. The primary eQTL/Gasperini results (§5.16) use the GenomicsxAI PyTorch port (genomicsxai/alphagenome-pytorch) with weights converted from the DeepMind JAX checkpoint. The GTEx and multi-tissue arms (§5.13, §5.20) use the **official DeepMind-served AlphaGenome API** (alphagenome==0.7.0, GeneMaskLFCScorer(RNA_SEQ), 1 Mb context); the two agree at r=0.990 on the shared eQTL set (§5.16 provenance). Before scoring, the port was validated locally: the GAPDH-TSS RNA-seq peak ratio is 6544× (SANITY_PASS) and the port authors report r=0.9999 track parity with the reference implementation. Variants are scored at the full 1,048,576 bp context with the exon-mask log-fold-change scorer (GeneMaskLFCScorer(RNA_SEQ)) over LCL RNA-seq output tracks (n=8) for eQTL, and by element ablation for Gasperini CRISPRi (22_run_alphagenome.py, 25_run_alphagenome_gasperini.py).

Evo2 (genomic language model)

Evo2-7B (evo2_7b, StripedHyena2 autoregressive genomic language model, no functional-genomics supervision) scores each variant zero-shot as the reverse-complement-averaged change in mean per-token log-likelihood over an 8,192 bp window centered on the variant (23_run_evo2.py, 24_score_evo2.py). The model fits on a single A100-40GB (≈22 GB VRAM).

scooby (single-cell decoder)

scooby (johahi/neurips-scooby, trained on the NeurIPS bone-marrow multiome) is the only oracle that decodes single-cell profiles, so it is evaluated on the single-cell distribution axis. Because no K562 scooby exists, K562 enhancer hits are scored via cross-cell-type transfer through the erythroid lineage (Proerythroblast/Erythroblast/Normoblast). For each hit the enhancer is ablated and the per-cell predicted expression shift is compared to the measured shift (energy distance vs a sampling null, and per-cell log2 fold change; 24_run_scooby.py, 25_score_scooby.py).

Level-playing-field oracle comparison (matched context and readout)

Because the headline eQTL comparison places oracles of very different design side by side, we audited whether the ranking could be an artifact of unequal deployment (different input context windows, readout tracks, or scoring effort) rather than a property of the models. All three eQTL oracles were run with an identical perturbation (single-base ref→alt swap) on an identical common variant set (n=1,681, verified identical across Borzoi∩AlphaGenome∩Evo2), scored against an identical label (fine-mapped effect-size tercile) with identical baselines.

Table F1 — audited deployment of each oracle.

Oracle	Architecture	Context (bp)	Readout	Tracks	eQTL AUROC [95% CI]
Borzoi	conv U-net (supervised coverage)	524,288	RNA-seq at TSS bin (±5)	28 LCL	0.600 [0.562–0.630]
AlphaGenome	conv+transformer (supervised, 2026)	1,048,576	gene-mask exon LFC	8 LCL	0.624 [0.587–0.657]
Evo2 (7B)	autoregressive genomic LM	8,192	per-token log-likelihood	0	0.516 [0.485–0.552]

Common-set baselines: distance-to-TSS 0.573, fine-mapping PIP 0.688. Two deployment differences could in principle bias the ranking, and both are controlled:

1. **Context window.** The concern is that AlphaGenome's higher AUROC reflects its 2× larger receptive field. We rule this out by a *geometric bound*: the common set was constructed under Borzoi's 524,288 bp geometry, and the maximum variant-to-TSS distance in it is **259,712 bp, smaller than Borzoi's 262,144 bp half-window** (median 9.4 kb). Every scored pair therefore has both endpoints inside Borzoi's window; AlphaGenome's extra context adds only distal flanking sequence, never additional variant↔gene span. The gap cannot be a coverage artifact. We confirmed this empirically by re-running AlphaGenome at Borzoi's exact 524,288 bp context on the same common set: the AUROC is **0.626 [0.592–0.658]**, statistically identical to the 1,048,576 bp run (0.624; DeLong z=0.28, p=0.78) and unchanged relative to Borzoi (0.600). Halving AlphaGenome's context to match Borzoi's leaves its score essentially identical, so AlphaGenome's small point-estimate edge, itself not significant (next paragraph), does not come from its larger context window. Moreover the gap is not significant to begin with: on the identical common set a DeLong test finds AlphaGenome (0.624) and Borzoi (0.600) indistinguishable (z=1.75, p=0.081), and both are out-ranked by the PIP prior.
2. **Readout track count (28 Borzoi vs 8 AlphaGenome vs 0 Evo2).** Track count is a fixed property of each released head, not a tunable. Borzoi's *larger* LCL panel (28 vs 8) works against AlphaGenome, so it cannot explain AlphaGenome's ranking above Borzoi. Evo2 has no cell-type readout, and its 8 kb window reaches the TSS for only 37.4% of common-set variants, so its near-chance 0.516 is conservative.

Residual asymmetries are stated honestly: the three oracles use their respective recommended variant-effect readouts (forcing one readout would deploy models off-label), and the AlphaGenome weights were validated against DeepMind's official API (r=0.990; §5.16). None of these asymmetries favors the headline conclusion, so the central negative result is robust to a level-playing-field deployment.

Statistical rigor

Every headline discrimination AUROC carries a 95% percentile bootstrap CI (2000 resamples, stratified within-class, seed 0). Oracle-vs-baseline comparisons use the fast midrank DeLong test (Sun & Xu 2014, two-sided) on matched samples (the oracle's in-window subset), and p-values are corrected by Benjamini-Hochberg FDR over the comparison family (`apply_rigor.py`, `stats.py`). Point estimates match the committed per-track scorers.

Code and frozen artifacts (19 score JSONs, 20 figures, all scripts): **github.com/Qihao-Duan/PerturbExpression**.

A living benchmark: submission format, scoring harness, and leaderboard

To make CIS-Δ a resource others can submit to rather than a one-off evaluation, we release three coupled components. (i) A documented **submission format** (`SUBMISSION_FORMAT.md`): per axis, a table keyed by the axis's `pair_id` carrying a predicted expression change `pred_delta` (and optional confidence), with unscored pairs encoded as missing so coverage is reported explicitly and never silently imputed. (ii) A **frozen ground-truth bundle** (`results_committed/frozen_truth/`): one self-contained table per axis with the endogenous labels and both reference baselines (distance-to-TSS, fine-mapping PIP), so a submission is scored from the committed repository without access to the raw single-cell/eQTL data. (iii) A **scoring harness** (`code/score_submission.py`) that recomputes an axis's metrics (AUROC with stratified bootstrap CIs, sign-agreement, Spearman, and DeLong tests against the baselines with Benjamini–Hochberg control), reusing the same `stats.py` module as the primary analyses; its `--selftest` round-trips the committed oracle predictions and asserts recomputed values match published scores to <1e-6. A static leaderboard (`docs/leaderboard.html`, `docs/leaderboard.md`, generated by `code/gen_leaderboard.py`) ranks every model against the baselines per axis, so the central calibration/coverage gap is visible at a glance and updates as new models are submitted. We note the governance status of these components: they are a working infrastructure prototype, and the accompanying `SUBMISSION_FORMAT.md` and leaderboard carry an explicit provisional banner. The datasets are held at pre-admission status and the leaderboard is not published as a scientific resource; both are gated on the project's independent go/no-go review (Gate A route selection, then Gate C release authorization). The harness, format, and self-test are stable and runnable today; what is deferred is the *public-release* claim, not the code.

Data and code availability

- **Dataset 1 (anchor, K562 CRISPRi):** GEO **GSE120861** (Gasperini et al. 2019), build hg19/GRCh37 — main finding + single-cell axis.
- **Dataset 2 (cross-cell-type CRISPRi):** EngreitzLab `EPCrisprBenchmark` held-out set (Gschwind et al. 2023, ENCODE-rE2G), build GRCh38 — WTC11, HCT116, Jurkat, GM12878; github.com/EngreitzLab/CRISPR_comparison.
- **Dataset 3 (CRISPRa, up-regulation):** SRA **PRJNA1157910** (Chardon et al. 2024), build hg38 — K562 + iPSC-neurons.
- **Dataset 4 (natural variation, cis-eQTL):** eQTL Catalogue **QTD000110** (GEUVADIS LCL, SuSiE fine-mapped, n = 445), build hg38 — 1,826 high-confidence (PIP > 0.5) single-nucleotide lead variants across 1,595 genes; the CIS-Δ natural-variation axis (§5.11).
- **Dataset 5 (natural variation, second cohort + cell-type specificity):** eQTL Catalogue GTEx v10 (study QTS000015, SuSiE fine-mapped, hg38) — whole blood **QTD000356** (n = 853; §5.13), plus brain cortex **QTD000171**, skeletal muscle **QTD000281**, and sun-exposed skin **QTD000316** as the four-tissue cell-type- specificity panel (§5.20).
- **Dataset 6 (reporter variant effect, bidirectional single-base MPRA):** GEO **GSE126550** (Kircher et al. 2019, *Nat Commun* 10:3583), builds GRCh38 + GRCh37 — saturation mutagenesis of 20 disease regulatory elements (episomal MPRA reporter); the reporter variant-effect axis (§5.21). Per-variant

effects **reconstructed** from the deposited raw DNA/RNA tag-count tables using the authors' published model (the authors' effect table is OSF-hosted and was not retrievable); labeled a reconstruction throughout.

- **Dataset 7 (GWAS-variant single-cell CRISPRi):** GEO **GSE171452** (Morris et al. 2023, *Science* 380:eadh7699, STING-seq), build hg19 — K562 GWAS-cCRE single-cell CRISPRi; the cross-screen-concordance axis (§5.22), published significant-links subset (Supplementary Table S3F).
- Reference genomes: hg19 (datasets 1, 7), hg38 (datasets 2–5), GRCh38+GRCh37 (dataset 6).
- Oracle weights: `johahi/borzoi-replicate-0` (borzoi-pytorch 0.5.1), `EleutherAI/enformer-official-rough` (enformer-pytorch 0.8.12).
- Benchmark code, splits, frozen scores, and all figures: **github.com/Qihao-Duan/PerturbExpression** — every reported number traces to a committed JSON; the WORKLOG records each dataset-build step for reproducibility.

Author contributions /acknowledgements

(To be completed by the authors.)

References

Provenance tags: **[R]** retrieved and confirmed in this project's literature searches /live database checks; **[V]** to verify against primary source before final submission (cited from background knowledge).

Datasets used (seven scoring fixtures: five legacy established tracks and two externally inspected provisional tracks)

1. **[R]** Gasperini M. et al. *A genome-wide framework for mapping gene regulation via cellular genetic screens.* **Cell** 176(1–2):377–390 (2019). GEO: **GSE120861** (verified live against NCBI: *Homo sapiens*, 53 samples, build hg19). — the CIS-Δ v1 anchor (K562 CRISPRi; deep single-cell axis).

1b. **[R]** Gschwind A.R., Mualim K.S., Karbalayghareh A., Sheth M.U. et al. *An encyclopedia of enhancer-gene regulatory interactions in the human genome.* **bioRxiv** 2023.11.09.563812 (ENCODE-rE2G). Curated CRISPR benchmark of 10,411 element–gene pairs; the `EPCrisprBenchmark` held-out non-K562 subset (WTC11, HCT116, Jurkat, GM12878; GRCh38) is the CIS-Δ cross-cell-type replication set. Repo: github.com/EngreitzLab/CRISPR_comparison.

1c. **[R]** Chardon F.M., McDiarmid T.A. et al. *Multiplex, single-cell CRISPRa screening for cell type specific regulatory elements.* **Nat Commun** 15:8209 (2024). DOI 10.1038/s41467-024-52490-4. SRA **PRJNA1157910**. K562 + iPSC-neuron CRISPR-activation; the CIS-Δ up-regulation (positive-Δ) dataset.

1d. **[R]** eQTL Catalogue (Kerimov N. et al., *A compendium of uniformly processed human gene expression and splicing quantitative trait loci*, Nat Genet 2021) — GEUVADIS LCL cis-eQTL, dataset **QTD000110**, SuSiE fine-mapped, build hg38 (n = 445). 1,826 high-confidence (PIP > 0.5) single-nucleotide lead variants; the CIS-Δ natural-variation axis (§5.11).

1e. **[R]** eQTL Catalogue — GTEx v10 whole blood cis-eQTL, dataset **QTD000356**, SuSiE fine-mapped, build hg38 (n = 853 donors). 1,800 fine-mapped lead variants; the second natural-variation cohort (§5.13).

1f. **[R]** Kircher M., Xiong C., Martin B. ... Shendure J., Ahituv N. *Saturation mutagenesis of disease-associated regulatory elements.* **Nat Commun** 10:3583 (2019). DOI 10.1038/s41467-019-11526-w. GEO **GSE126550**. Bidirectional single-base MPRA reporter over 20 regulatory elements; the reporter variant-effect axis (§5.21). Effects reconstructed from deposited raw counts (authors' model); coordinate concordance vs the paper 0.97–1.00, SORT1 C/EBP footprint recovered.

1g. **[R]** Morris J.A., Caragine C., Daniloski Z. ... Sanjana N.E. *Discovery of target genes and pathways at GWAS loci by pooled single-cell CRISPR screens (STING-seq).* **Science** 380:eadh7699 (2023). GEO **GSE171452**. GWAS-variant single-cell CRISPRi in K562; the cross-screen-concordance axis (§5.22), published significant-links subset (168 CRE→gene links /124 genes, hg19).

Therapeutic motivation (Casgevy /BCL11A enhancer)

2. **[R]** Frangoul H. et al. *Exagamglogene autotemcel for severe sickle cell disease.* **NEJM** — DOI 10.1056/NEJMoa2309676.

3. **[R]** Frangoul H. et al. *CRISPR-Cas9 gene editing for sickle cell disease and β-thalassemia* (first-in-human). **NEJM** — DOI 10.1056/NEJMoa2031054.

4. **[R]** Casgevy /exa-cel FDA-approval review. PMC11305803.

Sequence-to-expression oracles (benchmarked /discussed)

5. **[R]** Avsec Ž. et al. *Effective gene-expression prediction from sequence by integrating long-range interactions* (**Enformer**). Nat Methods 18:1196–1203 (2021). Weights: `EleutherAI/enformer-official-rough` (enformer-pytorch 0.8.12).

6. **[R]** Linder J. et al. *Predicting RNA-seq coverage from DNA sequence at single-base resolution* (**Borzoi**). Nat Genet (2025). Weights: `johahi/borzoi-replicate-0` (borzoi-pytorch 0.5.1).

7. **[R]** scooby — single-cell-resolution profiles from sequence. PMC11429888 /PMC12615262.

8. **[R]** Avsec Ž., Latysheva N., Cheng J., Novati G. et al. *Advancing regulatory variant effect prediction with AlphaGenome.* **Nature** 649, 1206–1218 (2026). DOI 10.1038/s41586-025-10014-0 (preprint bioRxiv 2025.06.25.661532). Weights/code: github.com/google-deepmind/alphagenome_research; benchmarked here via the community PyTorch port (§5.16, Methods).

Enhancer–gene link models (baselines)

9. **[R]** Fulco C.P., Nasser J., Jones T.R., Munson G. et al. *Activity-by-contact model of enhancer–promoter regulation from thousands of CRISPR perturbations.* **Nat. Genet.** 51, 1664–1669 (2019). DOI 10.1038/s41588-019-0538-0.

10. **[R]** ENCODE-rE2G — supervised enhancer–gene regulatory link model; the genome-wide DNase-only predictions used here are the ENCODE release accompanying Gschwind A.R. et al. (ref. 1b, bioRxiv 2023.11.09.563812).

11. **[R]** Jiang F. et al. *EPInformer: scalable and integrative prediction of gene expression from promoter–enhancer sequences with multimodal epigenomic profiles.* **Nat. Commun.** (2026), article s41467-026-70535-8 (preprint bioRxiv 2024.08.01.606099). Repo: github.com/pinellolab/EPInformer.

Perturbation benchmark culture

12. **[R]** Arc Virtual Cell Challenge — trans-perturbation prediction benchmark.

Generative cis-regulatory design (Track B corpora)

13. **[R]** DNA-Diffusion. Nat Genet s41588-025-02441-6.
14. **[R]** D3 (discrete diffusion for regulatory design). PMC13142467.
15. **[R]** SPICE (splicing-element design). PMC12637576.
16. **[R]** AAV /cell-type-specific promoter design. bioRxiv 2026.01.13.699371.
17. **[V]** Gosai S. /Sabeti P. — synthetic-enhancer lentiMPRA sets.

Perturbation-response tracks (breadth resource)

23b. **[R]** Morris J.A., Caragine C., Daniloski Z. ... Sanjana N.E. *Discovery of target genes and pathways at GWAS loci by pooled single-cell CRISPR screens (STING-seq).* Science 380 (2023), eadh7699; GEO GSE171452. GWAS-variant single-cell CRISPRi in K562; the cross-screen-concordance track (§5.22).

24b. **[R]** Kircher M., Xiong C., Martin B. ... Shendure J., Ahituv N. *Saturation mutagenesis of twenty disease-associated regulatory elements at single base-pair resolution.* Nature Communications 10 (2019), s41467-019-11526-w; GEO GSE126550. Bidirectional single-base MPRA reporter; the reporter variant-effect axis (§5.21). Effects reconstructed from deposited raw counts using the authors' published model.

CRISPRi validity /concordance

18. **[R]** CRISPRi vs. eQTL effect-size concordance. bioRxiv 2025.05.05.651929 ($r \approx 0.88$–0.92 for well-powered CREs).
19. **[R]** TAP-seq. Nat Methods s41592-020-0837-5 (accession GSE135497 — [V]).
20. **[R]** Genome-scale hPSC CRISPRi map. Cell Genomics S2666-979X(25)00332-5.

Population /fine-mapped eQTL (Track A natural-variation axis)

21. **[R]** Concurrent work — *Practical utility of sequence-to-omics models for fine-mapping.* bioRxiv (Feb 2026). Evaluates AlphaGenome/Borzoi/Enformer/Sei variant-effect predictions on GTEx/TOPMed/MAGE fine-mapped eQTLs; the closest prior/concurrent evaluation to the CIS-Δ natural-variation axis (§5.11). Our distinct contribution is the unified benchmark unit + scoring axes shared with the CRISPRi/CRISPRa tracks, not first-ever eQTL scoring.

22. **[V]** OneK1K; Cuomo et al.; Jerber et al. — sc-eQTL cohorts (future single-cell natural-variation extension).

Reference cell atlases /pretraining contexts (future work)

23. **[V]** Arc Virtual Cell Atlas; scBaseCount; Tahoe-100M; X-Atlas/Orion.